

# Lead-biased Grounding Score: injecting discourse-structural knowledge into reference-free, cost-constrained assessment of summary quality

Mikolaj Semeniuk<sup>a</sup>, Agnieszka Jastrzebska<sup>a,\*</sup>

<sup>a</sup>*Faculty of Mathematics and Information Science, Warsaw University of  
Technology, Koszykowa 75, Warsaw, 00-662, Poland*

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## Abstract

Assessing the quality of machine-generated summaries is a decision problem that deployed systems must solve continuously: which model version to release, which candidate to serve, and when to block a regression. The metrics that correlate best with human judgment are unusable in that setting, because they require either a human-written reference summary or a large language-model judge call per candidate. We introduce the Lead-biased Grounding Score, a reference-free evaluation operator that injects one explicit piece of domain knowledge — the journalistic convention that salient content concentrates in the opening sentences of a news article — as a parametric prior over source-sentence positions, and combines it with sentence-level grounding in a small sentence-embedding space. The operator has a single interpretable hyperparameter and needs one embedder forward pass per sentence: no reference, no judge, no fine-tuned encoder. We evaluate fourteen metrics on the SummEval benchmark under one controlled hardware budget, with wall-clock runtime measured for every metric. Our operator attains summary-level Spearman correlations of 0.396, 0.227, 0.156 and 0.356 on coherence, consistency, fluency and relevance. Against thirteen published baselines it strictly dominates six widely used metrics on the cost-quality plane, including BERTScore, whose mean correlation it exceeds while running ten times faster. Against trivial lead-window controls we build in re-

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\*Corresponding author.

*Email addresses:* m.semeniuk@icloud.com (Mikolaj Semeniuk),  
A.Jastrzebska@mini.pw.edu.pl (Agnieszka Jastrzebska)

sponse to reviewer feedback — keep the first  $k$  source sentences and match them lexically, no model at all — LGS is no longer Pareto-optimal: the cheapest lead-window control matches or beats its raw mean correlation at roughly three orders of magnitude lower cost. Investigating why implicates the benchmark, not only the metric: SummEval’s human ratings are substantially confounded with how much a candidate copies verbatim from the source, and a zero-parameter bigram copy-rate counter outranks eleven of the fourteen published metrics in our pool on this axis. Once that confound is partialled out, LGS is no longer statistically distinguishable from the cheapest lead-window control, though it ties BERTScore. Re-running the entire selection procedure inside a cluster bootstrap further shows that the knowledge prior is worth switching on — the no-prior setting is selected in under 1% of resamples — but that its exponent is not identifiable at benchmark scale: the reported value is selected in 18% of resamples, and an unchanged backbone scored on two draws of the same 50 articles agrees with itself on the exponent only 44% of the time. We therefore present the prior as a design element carrying an obligatory, and demonstrably underdetermined, calibration step.

*Keywords:* Summarization quality assessment, reference-free evaluation, knowledge-informed prior, sentence embeddings, cost-aware decision support, hyperparameter calibration

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## 1. Introduction

Every organisation that operates a summarisation service has to answer the same question many times a day: *is this summary good enough?* The question arises when a new model checkpoint is proposed for release, when several decoding configurations compete for the same request, when a nightly regression suite must decide whether a change degraded output quality, and when a reinforcement-learning loop needs a scalar reward for a freshly sampled candidate. In all of these situations the answer is consumed by an automated decision procedure, not by a human reader, and it must be produced quickly, deterministically, and without any artefact that a production pipeline does not have.

The evaluation metrics that agree best with human judgment fail exactly this test. Reference-based metrics, including BLEU [1], ROUGE-L [2], METEOR [3], ChrF [4], BERTScore [5], MoverScore [6], BARTScore [7] and

SMART [8], all presuppose a human-written reference summary — a precondition that production inputs never satisfy, because the candidate is the only summary in the system. The two strongest *reference-free* contenders on the SUMMEVAL [9] benchmark avoid that dependency but replace it with an inference cost that dominates the engineering budget: UniEval requires a fine-tuned T5-large pass per (article, candidate, dimension) tuple [10], and G-Eval queries a large language-model judge with a chain-of-thought prompt and aggregates over multiple sampled generations [11]. Recent surveys of reference-free text-generation metrics [12] and of the LLM-as-a-judge paradigm [13] document the same tension: judge-based evaluation buys correlation with human raters at the price of latency, monetary cost, non-determinism and reproducibility.

There is, however, a resource that deployed summarisation pipelines do possess and that current metrics leave unused: *explicit knowledge about how source documents are structured*. News writing follows the inverted-pyramid convention, in which the most newsworthy material is placed in the opening sentences; the resulting *lead bias* of news corpora is one of the best-documented regularities in the summarisation literature [14, 15]. This knowledge costs no extra model forward pass at inference time — applying it is a closed-form arithmetic reweighting of a similarity already computed, not an additional model, annotation or retrieval step — and it is exactly the kind of structural domain regularity that a knowledge-informed system should encode explicitly rather than hope a learned representation will rediscover.

This paper makes the following contributions.

1. **A knowledge-informed evaluation operator.** We define the Lead-biased Grounding Score (LGS), which combines sentence-level grounding of the candidate in the source with an explicit, one-parameter prior over source-sentence position that encodes discourse-structural knowledge about news documents (Section 3). The operator is reference-free, interpretable — every candidate sentence is anchored to one identifiable source sentence — and runs at one small-embedder forward pass per sentence, with no judge call and no fine-tuned encoder.
2. **A cost-controlled benchmark of fourteen metrics.** We reproduce thirteen published metrics plus LGS on SUMMEVAL under one hardware budget and one runtime harness, and report wall-clock cost per sample alongside correlation for every metric (Section 6). To our knowledge, no prior meta-evaluation of summarisation metrics reports

measured deployment cost and correlation on the same axis for a baseline pool of this size.

3. **A Pareto analysis of the cost–quality plane, including trivial lead-window controls, and the extractiveness confound this analysis exposes.** Among the fourteen published metrics, LGS strictly dominates six widely deployed metrics, among them BERTScore, which it exceeds in mean summary-level Spearman  $\rho$  while running roughly ten times faster, and the LLM-judge metric in our reproduction is itself dominated by the fine-tuned-encoder metric. But when the pool is extended with reference-free lead-window controls — keep the first  $k$  source sentences, match with ROUGE-L, no model — LGS is *not* Pareto-optimal: the cheapest control matches or exceeds its raw mean correlation at roughly  $950\times$  lower cost (Section 6.3). We trace this to a confound: SummEval’s human ratings substantially reward verbatim copying from the source, and a parameter-free copy-rate counter alone outranks eleven of the fourteen published metrics on mean correlation. Controlling for this confound, LGS is not statistically distinguishable from the cheapest lead-window control and ties BERTScore (Section 6.3). We report this as a limitation of the metric and, more importantly, of the benchmark axis itself.
4. **A measurement of how much of a tuned structural prior is signal and how much is the luck of the development draw.** Metric papers report the selected hyperparameter as a number; we instead re-run the *entire selection procedure* on every cluster-bootstrap resample and report the resulting distribution over grid values (Section 7.3). Two facts follow. Switching the prior on is robust:  $\lambda=0$  is selected in 0.7% of development resamples and 0.3% of test resamples. Which exponent to switch it to is not: the reported  $\lambda^*=0.5$  is selected in 17.8% of resamples, and a fixed backbone compared against itself on two independent draws of the same 50 articles agrees on the exponent only 43.5% of the time. We therefore present LGS as a design plus an obligatory calibration protocol (Section 4) whose output is a usable interval rather than a reproducible point, and we report the width of that interval instead of suppressing it.
5. **A reproducible artefact.** The full pipeline — all fourteen metrics, the calibration sweep, the backbone ablation and every table and figure in this paper — is released as open-source code with a single-command reproduction path at

<https://github.com/mikolajsemeniuk/llmbench>.

We are explicit from the outset about what is *not* claimed. LGS does not attain the highest correlation with human judgment in our pool: UniEval does, and we report that plainly. Nor is LGS Pareto-optimal once trivial lead-window controls enter the pool: a four-line lexical heuristic matches or beats it on raw mean correlation at far lower cost, and the two are not statistically distinguishable once the benchmark’s extractiveness confound is controlled for. The claim we defend is narrower: among metrics that perform any learned or embedding-based computation, LGS occupies a cost–quality point that none of them dominates, and it is the cheapest metric in the pool that ties BERTScore’s confound-controlled correlation.

The remainder of the paper is organised as follows. Section 2 surveys reference-based and reference-free metrics, organises them by the knowledge source they consume, and states the research gap. Section 3 defines LGS, gives its algorithm and computational complexity, and motivates each design choice. Section 4 presents the calibration protocol for the knowledge prior and explains why it is a necessary rather than a decorative step. Section 5 describes the experimental protocol. Section 6 reports the cost–quality analysis, including the Pareto frontier. Section 7 reports correlations with human judgment together with paired-bootstrap significance tests, the prior sweep, the backbone ablation, and the cross-backbone calibration result. Section 8 discusses deployment, limitations and threats to validity, and Section 9 concludes.

## 2. Related Work and Research Gap

### 2.1. *Reference-based metrics*

The longest-standing family of summarisation metrics is reference-based: a candidate summary is compared against one or more human-written reference summaries. N-gram precision and recall metrics (BLEU [1], ROUGE-L [2], METEOR [3], ChrF [4]) operate on surface-form overlap. Their well-documented blind spot is that paraphrases of equivalent content score low simply because the surface forms differ, which is precisely the regime abstractive summarisation operates in. Contextual-embedding metrics replace surface overlap with cosine similarity in a learned representation space derived from Transformer encoders [16, 17]: BERTScore [5] matches token embeddings greedily, MoverScore [6] solves a transportation problem over them,

BARTScore [7] re-uses the encoder–decoder log-likelihood under a pretrained generation model based on BART [18], and SMART [8] extends the matching to sentence units. A parallel line of work in machine translation evaluation learns metric weights directly against human judgments (BLEURT [19], COMET [20]), and on the summarisation side a related family targets faithfulness specifically (FactCC [21] for factual consistency classification, SummaC [22] for NLI-based inconsistency detection). They alleviate the paraphrase problem but inherit the same structural dependence on the reference: the score is a function of (reference, candidate) pairs and is undefined when no reference exists. Two recent studies sharpen the case for entailment-based rather than similarity-based factuality scoring: Ye et al. [23] show that aggregating several imperfect NLI-style consistency metrics through preference learning improves factual-consistency judgments over any single metric, and Mujahid et al. [24] stress-test six widely used reference-free factuality metrics under paraphrase-preserving perturbations and find that similarity-based scorers, unlike entailment-based ones, are frequently fooled by perturbations that change surface form without changing meaning — exactly the failure mode cosine similarity is exposed to.

## 2.2. Reference-free and source-based metrics

A smaller body of work scores candidates against the *source* article rather than against a human reference, removing the deployment dependency on hand-written summaries; a recent survey maps this family across generation tasks [12]. G-Eval [11] prompts a large language model [25] with a chain-of-thought rubric [26] to score each (source, candidate) pair on the relevant dimension. It obtains the strongest published correlation on SUMMEVAL’s consistency dimension among published metrics, but at a substantial cost: one or more LLM-judge calls per (article, candidate, dimension) tuple, with sampled decoding requiring repeated calls per pair to absorb LLM-judge variance (the original paper averages 20 GPT-4 samples). Surveys of the judge paradigm document the accompanying issues of prompt sensitivity, length bias, non-determinism and dependence on proprietary, frequently-updated endpoints [13], and a growing line of work explicitly targets cheaper substitutes for judge calls in evaluation [27]. The metric is therefore inappropriate for deployment scenarios that require deterministic, low-latency, high-throughput scoring. UniEval [10] is a hybrid: a fine-tuned T5-large encoder [28] is queried with a Boolean question (“Is this summary {coherent, consistent, fluent, relevant}?”) and the score is the posterior probability of

“Yes”. UniEval sidesteps the LLM-judge cost but still requires a fine-tuned encoder per dimension and a forward pass per (article, candidate, dimension) tuple. GPTScore [29] re-uses a pretrained language model to compute a per-token log-likelihood of the candidate conditioned on the source, requiring a single forward pass per pair but inheriting the model’s calibration as the score’s calibration.

A pre-LLM family of reference-free, source-based metrics is the closest prior art to LGS and must be positioned explicitly. SUPERT [30] builds a *pseudo-reference* by extracting salient source sentences and then scores the candidate against that pseudo-reference with contextual-embedding similarity; ESTIME [31] uses BERT embeddings to flag points of semantic mismatch between summary and source directly, without an intermediate reference; BLANC [32] measures how much a summary improves a masked-language model’s ability to reconstruct held-out source tokens; and SummaQA [33] checks whether a QA model can answer source-derived cloze questions using only the summary. SUPERT and ESTIME are mechanistically the closest antecedents to LGS: SUPERT already performs reference-free, salience-sensitive, embedding-based scoring against the source, and ESTIME already performs embedding-level grounding checks between summary and source without an intermediate reference. LGS differs from this family in two specific ways rather than in kind: (a) it grounds each *candidate sentence* directly against raw source sentences via mean-of-max cosine, instead of comparing against an intermediate pseudo-reference (SUPERT) or an auxiliary masked-LM/QA task (BLANC, SummaQA); and (b) it encodes positional lead bias as an explicit, tunable multiplicative prior inside the scoring function, rather than leaving position implicit in an extraction step. The contribution claimed below is this specific combination — direct sentence-level grounding plus an explicit, calibrated lead-position prior, evaluated jointly against deployment cost — not the introduction of reference-free source-grounded evaluation as a category, which this family already established.

### 2.3. The closest control: whole-text embedding cosine

The simplest reference-free embedding baseline is a single cosine similarity between the embedding of the entire source and the embedding of the entire candidate, computed with the same sentence-embedder used by the metric under test. We call this baseline EMBEDSCORER. Two properties make it ideal for like-for-like comparison with LGS. First, it shares the embedder backbone, so any difference between the two metrics is attributable to

the architectural choices of LGS — sentence-level grounding via  $\max_i$  on the source side, mean aggregation on the candidate side, and the position prior  $w(i)$  — and not to a stronger or weaker embedding model. Second, both metrics are reference-free; the comparison isolates the architectural contribution from the orthogonal question of whether reference-based metrics are stronger than reference-free ones in general. We return to the EMBEDSCORER-vs-LGS gap as the controlled ablation of the knowledge component in Section 7.

#### 2.4. Research gap

Table 1 organises the metric pool *benchmarked in this paper* by the knowledge sources each metric consumes at inference time; SUPERT, BLANC, SummaQA and ESTIME are discussed above but are not run in our controlled-budget harness and so are not rows of this table. Within the benchmarked pool, no metric consumes *a priori knowledge about the structure of the source document* as an explicit, inspectable scoring component, even though such knowledge costs no extra forward pass to apply and is one of the most robust regularities in the news domain that the dominant benchmarks are built on. SUPERT is the closest published exception, and it encodes source structure only implicitly, through which sentences its extractive pseudo-reference step happens to select, not as a separately calibrated parameter.

Prior systems do exploit lead bias, but only *implicitly* and only on the generation side: the lead-3 extractive baseline is a standing strong competitor on CNN/DailyMail, and pretrained encoder-decoder summarisers absorb the same prior through their training distribution [34, 35, 36]. On the *evaluation* side, published reference-free metrics either leave the prior implicit (SUPERT’s pseudo-reference extraction) or do not use it at all. LGS extends this line by making the structural prior an explicit, inspectable, single-parameter component of the scoring function — and, because it is explicit, a component that can be ablated, swept, and shown to be necessary or not, which an implicitly-absorbed prior cannot.

A second gap concerns how metric papers are evaluated. Published meta-evaluations report correlation with human judgment but not measured deployment cost on the same axis for a comparable-sized baseline pool; Meta-Metrics [37], for instance, benchmarks thirteen summarisation metrics but reports only a parameter count for one reward-model variant, not runtime, and only for that variant. Without a cost axis one cannot tell whether a metric is a sensible deployment choice or merely a strong one, and one cannot

detect metrics that are dominated — slower *and* weaker than an available alternative. Section 6 supplies that axis under a single controlled budget and shows that domination is common in the current pool.

Each column below is defined independently of LGS, so the table can be read without circularity: *Ref.* is whether the metric’s score is a function of a human-written reference summary (undefined without one); *Src.* is whether it reads the source article at all; *Model* is whether scoring requires a pretrained or fine-tuned neural network (a lexical metric needs none); *Judge* is whether an autoregressive LLM is prompted to produce the rating directly, rather than a fixed scoring function consuming a representation; *Prior* is whether document-position information is an explicit, separately inspectable and tunable input to the scoring function, as opposed to being absorbed implicitly by a learned representation or by an upstream extraction step (Section 2.4 discusses SUPERT, the one published metric that uses source structure implicitly). LGS’s own row follows directly from these definitions rather than being asserted: it has no Ref. tick because Equation 1 never receives a reference summary as input, and no Judge tick because scoring is a fixed cosine-and-argmax computation with no LLM call anywhere in it — the same reason G-Eval is the only Judge row and UniEval, despite consuming both source and reference, is not, since its score is a fine-tuned encoder’s posterior rather than a prompted judgment. A blank cell means the column’s resource is absent from that metric’s inputs, not merely unreported.

Table 1: The metric pool organised by the knowledge sources consumed at inference time; see the paragraph above for how each column is defined and how to read a blank cell.

| Metric                      | Family             | Ref. | Src. | Model | Judge | Prior |
|-----------------------------|--------------------|------|------|-------|-------|-------|
| BLEU, ROUGE-L, ChrF, METEOR | lexical overlap    | ✓    | —    | —     | —     | —     |
| SMART-String                | lexical, sentence  | ✓    | —    | —     | —     | —     |
| EMBEDSCORER                 | embedding, doc.    | ✓    | —    | ✓     | —     | —     |
| SMART-Model                 | embedding, sent.   | ✓    | —    | ✓     | —     | —     |
| BERTScore                   | embedding, token   | ✓    | —    | ✓     | —     | —     |
| MoverScore                  | embedding, transp. | ✓    | —    | ✓     | —     | —     |
| BARTScore, GPTScore         | likelihood         | ✓    | —    | ✓     | —     | —     |
| UniEval                     | fine-tuned QA      | ✓    | ✓    | ✓     | —     | —     |
| G-Eval                      | LLM judge          | —    | ✓    | ✓     | ✓     | —     |
| <b>LGS (this work)</b>      | embedding, sent.   | —    | ✓    | ✓     | —     | ✓     |

### 3. The Lead-biased Grounding Score

#### 3.1. Notation

Table 2 collects the symbols used in the definition below and reused throughout Sections 3–7.

#### 3.2. Definition

Let  $D$  be a source document split into  $n$  sentences  $s_0, \dots, s_{n-1}$  and let  $C$  be a candidate summary split into  $m$  sentences  $c_0, \dots, c_{m-1}$ . Let  $\text{emb}(\cdot)$  be a sentence embedder of the Sentence-BERT family [38] (in the empirical part, we use `nomic-embed-text` [39] served via Ollama, see Section 5). The Lead-biased Grounding Score is defined as

$$\text{LGS}(D, C) = \frac{1}{m} \sum_{j=0}^{m-1} \max_{0 \leq i < n} w(i) \cdot \cos(\text{emb}(c_j), \text{emb}(s_i)), \quad (1)$$

where the source-position prior is

$$w(i) = \exp(-\lambda \cdot i/n), \quad (2)$$

Table 2: Notation used to define LGS and its calibration protocol.

| Symbol                     | Meaning                                                  |
|----------------------------|----------------------------------------------------------|
| $D$                        | Source document                                          |
| $C$                        | Candidate summary                                        |
| $n, m$                     | Number of source / candidate sentences                   |
| $s_i, c_j$                 | $i$ -th source sentence, $j$ -th candidate sentence      |
| $\text{emb}(\cdot)$        | Sentence embedder (Sentence-BERT family)                 |
| $\cos(\cdot, \cdot)$       | Cosine similarity between two embeddings                 |
| $w(i)$                     | Source-position (lead-bias) prior weight                 |
| $\lambda$                  | Lead-bias exponent, the metric’s one hyperparameter      |
| $\lambda^*$                | Dev-selected value of $\lambda$                          |
| $\hat{\lambda}$            | Dev-argmax $\lambda$ for a single dimension (Table 12)   |
| $\rho$                     | Spearman rank correlation with human ratings             |
| $\bar{\rho}$               | Mean $\rho$ across the four SUMMEVAL dimensions          |
| $\bar{\rho}_{\text{part}}$ | $\bar{\rho}$ with copy rate partialled out (Section 6.4) |

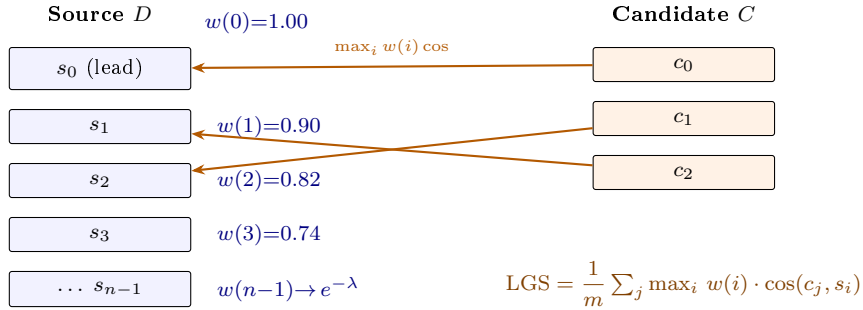


Figure 1: LGS operator. For each candidate sentence  $c_j$ , the metric finds the source sentence  $s_i$  that maximises the position-weighted cosine similarity  $w(i) \cdot \cos(c_j, s_i)$ , where  $w(i) = \exp(-\lambda i/n)$  is an exponential prior favouring the lead. The score is the mean of these per-candidate maxima. Orange arrows indicate illustrative argmax matches.

$i = 0$  denotes the lead sentence, and  $\lambda \geq 0$  is the only hyperparameter.  $\lambda = 0$  disables the prior and reduces LGS to position-agnostic mean-of-max recall. Figure 1 illustrates the operator graphically.

### 3.3. Algorithm and computational complexity

Algorithm 1 states the procedure. Every step is deterministic given the embedder, and the argmax index recorded in line 10 is the interpretability handle: it names the source sentence that the operator considers to be the

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**Algorithm 1** Lead-biased Grounding Score

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**Require:** source document  $D$ , candidate summary  $C$ , exponent  $\lambda \geq 0$ , sentence embedder  $\text{emb}(\cdot)$

**Ensure:** score in  $[0, 1]$  (negative cosine similarities are clamped to 0), plus per-sentence evidence anchors

```
1:  $(s_0, \dots, s_{n-1}) \leftarrow \text{SPLITSENTENCES}(D)$ 
2:  $(c_0, \dots, c_{m-1}) \leftarrow \text{SPLITSENTENCES}(C)$ 
3:  $E_i^s \leftarrow \text{emb}(s_i)$  for  $i = 0, \dots, n-1$   $\triangleright$  cached per source document
4:  $E_j^c \leftarrow \text{emb}(c_j)$  for  $j = 0, \dots, m-1$ 
5: for  $i \leftarrow 0$  to  $n-1$  do
6:    $w_i \leftarrow \exp(-\lambda \cdot i/n)$   $\triangleright$  structural prior, length-normalised
7: end for
8: total  $\leftarrow 0$ 
9: for  $j \leftarrow 0$  to  $m-1$  do
10:   $a_j \leftarrow \arg \max_{0 \leq i < n} w_i \cdot \cos(E_j^c, E_i^s)$   $\triangleright$  evidence anchor for  $c_j$ 
11:  total  $\leftarrow \text{total} + w_{a_j} \cdot \cos(E_j^c, E_{a_j}^s)$ 
12: end for
13: return  $\max(0, \text{total}/m), (a_0, \dots, a_{m-1})$ 
```

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evidence for each candidate sentence, so a low score can always be traced to the specific summary sentences that failed to find an anchor.

Let  $d$  be the embedding dimension and let  $T$  be the cost of one embedder forward pass. Embedding costs  $\mathcal{O}((n+m)T)$  and dominates in practice; the similarity stage costs  $\mathcal{O}(nmd)$  floating-point operations and the prior costs  $\mathcal{O}(n)$ . Memory is  $\mathcal{O}((n+m)d)$ , since the  $n \times m$  similarity matrix need never be materialised — the maximum over  $i$  can be accumulated in a single pass per candidate sentence.

Two properties matter for deployment. First, source embeddings depend only on  $D$ , so when  $K$  candidates are scored against the same article — the standard situation in model selection and in reinforcement-learning rollouts — the source cost is paid once and amortised, giving  $\mathcal{O}((n+Km)T)$  instead of  $\mathcal{O}(K(n+m)T)$ . Second, the exponent  $\lambda$  enters only through the weight vector  $w$ , so a full sweep over a  $\lambda$ -grid of size  $G$  reuses one set of embeddings and costs  $\mathcal{O}(Gnmd)$  additional arithmetic with no additional forward passes. Calibration (Section 4) is therefore essentially free relative to scoring, which is what makes an obligatory per-deployment sweep a realistic requirement rather than a burden.

For contrast, a judge-based metric costs one or more autoregressive decodes per (article, candidate, dimension) tuple, with the decode length itself dependent on the rubric; a fine-tuned-encoder metric costs one encoder pass per tuple, and thus scales linearly in the number of quality dimensions, which LGS does not.

### 3.4. Worked example

To make the operator concrete, consider a stylised article with  $n=5$  source sentences —  $s_0$ : “Apple announced a new iPhone at its September event.”;  $s_1$ : “The phone features an improved camera system.”;  $s_2$ : “Tim Cook presented the keynote in Cupertino.”;  $s_3$ : “Pre-orders begin Friday.”;  $s_4$ : “Stock rose 3% after the announcement.” — and a candidate summary with  $m=2$  sentences —  $c_0$ : “Apple unveiled a new iPhone.”;  $c_1$ : “Pricing starts at \$999.”. Suppose, for illustration purposes, that the cosine similarity matrix between the candidate sentence embeddings and the source sentence embeddings is

$$[\cos(c_j, s_i)]_{j,i} = \begin{pmatrix} 0.91 & 0.41 & 0.52 & 0.18 & 0.34 \\ 0.22 & 0.31 & 0.15 & 0.78 & 0.42 \end{pmatrix}.$$

At  $\lambda=0.5$  the position prior takes values  $w(0)\approx 1.00$ ,  $w(1)\approx 0.90$ ,  $w(2)\approx 0.82$ ,  $w(3)\approx 0.74$ ,  $w(4)\approx 0.67$ . For  $c_0$  the position-weighted argmax is  $s_0$  (weighted similarity  $1.00\cdot 0.91=0.91$ ); for  $c_1$  it is  $s_3$  ( $0.74\cdot 0.78=0.58$ ). The score is the mean of these two per-candidate maxima,  $\text{LGS}=(0.91+0.58)/2=0.745$ .

A hallucinated  $c_1$  that is unrelated to the source — say, “Tim Cook resigned as CEO.” with (assumed for illustration purposes) cosine row  $(0.18, 0.09, 0.55, 0.11, 0.21)$  — yields position-weighted maximum  $0.82\cdot 0.55=0.451$  on  $s_2$ , and the score drops to  $(0.91+0.45)/2=0.680$ . The metric’s sensitivity to ungrounded content is structurally tied to the candidate-side mean: with candidate length  $m$ , a single ungrounded sentence removes approximately  $1/m$  of the score’s mass.

### 3.5. Design rationale

*Sentence-level grounding.* A candidate sentence is *grounded* when some source sentence semantically supports it. The LGS metric (Eq. 1) implements this by taking, for each candidate sentence  $c_j$ , the maximum cosine similarity to any source sentence under the position prior, and then aggregating across the candidate with a mean. The asymmetry is deliberate:  $\max_i$  on the source side is a localised signal (“does this candidate

sentence have *any* anchor in the source?”);  $\text{mean}_j$  on the candidate side is a hallucination-sensitive aggregate (a single ungrounded candidate sentence depresses the score by  $1/m$ , where  $m$  is the candidate length, rather than being washed out by an averaging operator on both sides). We compared this asymmetric mean-of-max formulation empirically against alternative aggregations during the calibration phase — low-percentile recall, top- $k$  recall, candidate-side coverage exponent — and report the negative results in Appendix Appendix B; none lifted the development-set mean Spearman over the mean-of-max baseline.

*Which knowledge is injected, and why exponentially.* News summarisation corpora — and CNN/DailyMail in particular [40, 36], the source of all SUMM-EVAL articles — exhibit well-documented *lead bias* [14, 15]: salient information is disproportionately concentrated in the opening sentences of the article, a pattern reinforced by the inverted-pyramid convention of journalistic writing. This is the single piece of a priori domain knowledge that LGS encodes, and it enters the scoring function as a multiplicative source-side prior  $w(i) = \exp(-\lambda i/n)$  rather than being left for the embedder to infer. The exponential family is chosen for three reasons. First, parsimony: it is the simplest one-parameter decay shape that interpolates between a uniform prior ( $\lambda=0$ ) and a near-step prior ( $\lambda \rightarrow \infty$ ) without introducing a discontinuity. Second, scale-invariance: normalising the position  $i$  by the document length  $n$  makes the prior behave consistently across short and long articles, removing the need for a length-dependent tuning of  $\lambda$ . Third, multiplicative interaction: the prior reweights the cosine similarity *before* the  $\max_i$  operator, so it can move which source sentence wins the argmax for a given candidate; an additive prior would only shift scores uniformly without affecting the matching structure. The choice of an explicit prior over an implicit one is what makes the knowledge component *auditable*:  $\lambda=0$  removes it exactly, so its contribution can be measured rather than asserted (Section 7.2).

*Why a single hyperparameter.* The canonical LGS has exactly one tunable scalar,  $\lambda$ . Possible alternative formulations of this metric can be richer: a precision side mirroring BERTScore’s F-measure structure, a candidate-side coverage exponent that rewarded using more distinct source anchors, a within-summary diversity penalty, and a top- $k$  source aggregation. We tested each of these alternatives on the same held-out development split (Section 4) and rejected them for failing to lift the development-set mean Spearman

$\rho$  over the no-prior baseline. The rejected components and their negative numbers are reported in Appendix Appendix B. The single-hyperparameter design is the one we propose. A beneficial corollary of this discipline is that the calibration protocol has a small, well-defined search space (one scalar over a 1-D grid), which — combined with the embedding reuse noted in Section 3.3 — makes recalibration cheap enough to be mandatory.

*Relation to BERTScore.* The closest already-existing metric is BERTScore [5], which also computes a mean-of-max over cosine similarities between embedded units. However, there are three essential operational differences between LGS and BERTScore. First, BERTScore matches *tokens* (BPE or wordpiece) under a contextual encoder. In contrast, LGS matches *sentences* under a sentence encoder. The unit choice changes the semantic granularity: a token-level  $\max_i$  rewards lexical or stylistic overlap, while a sentence-level  $\max_i$  rewards proposition-level grounding and is structurally less sensitive to surface paraphrase. Second, BERTScore is reference-based: one side of every max is a human reference summary. LGS is reference-free: one side is the *source article*, removing the deployment dependency on human-written summaries (Section 2). The cost-quality implications differ accordingly: BERTScore’s correlation gain over n-gram metrics buys nothing in deployments where no reference exists. Third, neither BERTScore nor any other published mean-of-max metric uses a source-position prior. The lead-bias prior  $w(i)$  in LGS is the architectural ingredient that distinguishes it from a structurally simpler “sentence-BERTScore against the source” baseline; the EMBEDSCORER-vs-LGS comparison in Section 7 is the controlled experiment that isolates this contribution.

## 4. Calibrating the Knowledge Prior

### 4.1. Why calibration is a first-class step

A structural prior is a statement about the source documents, but it acts inside a learned representation space, and the two interact. Two separate findings make the sweep obligatory rather than decorative.

First, the point estimate of the optimum differs across backbones: with `nomic-embed-text` the development-set optimum is  $\lambda^* = 0.5$ , with `bge-m3` [41] it is  $\lambda^* = 0.25$ , and applying the nomic-tuned value to `bge-m3` *underperforms* the no-prior baseline (Section 7.5).

Second — and this is the finding that actually forces the protocol — the exponent is not identifiable from a 50-article development split at all, on any backbone. Section 7.3 measures this directly by re-running the selection procedure inside the cluster bootstrap: the selected value carries under a fifth of the selection mass, and an *unchanged* configuration disagrees with itself across two draws more than half the time. A practitioner who copies the published number is therefore not merely risking a backbone mismatch; they are copying one draw’s answer to a question that this much data cannot settle. What transfers is the sweep, which costs no additional embedder passes (Section 3.3), and the interval it returns.

#### 4.2. Article-level split

SUMMEVAL contains 100 source articles, each summarised by 16 systems, for 1,600 (source, candidate) pairs with human ratings on four dimensions (coherence, consistency, fluency, relevance). We split at the *article level*, deterministically by JSONL order: the first 50 articles form the development split, the remaining 50 the test split. The article-level split is critical because the 16 candidate summaries of a single article are not independent observations: a difficult article depresses correlations across all systems jointly. A naïve sample-level split would leak across the development/test boundary.

The JSONL-order split is deterministic and reproducible, but it is also an arbitrary 50/50 partition with no argument for why *this* partition, rather than another, should be the one that fixes  $\lambda^*$ . We test that directly: Table 3 re-runs the entire selection procedure — sweep the grid on a dev half, take the argmax, score it on the complementary test half — on 2,000 independent random 50/50 splits of the 100 articles, merging the per-sample scores already computed for the JSONL split (no additional embedder passes are needed). The value the JSONL split selects wins only 5.6% of random splits;  $\lambda = 0.25$  and  $\lambda = 0.3$  together account for 94.2% of the mass. This is the same instability documented from the resampling side in Section 7.3, now shown from the orthogonal angle of which 50 articles are used at all, and it argues against reading  $\lambda^*$  as a property of the metric rather than of one particular partition of SummEval.

#### 4.3. Selection criterion and verification

We sweep  $\lambda \in \{0, 0.25, 0.5, 1.0, 2.0\}$  on the development split, score each variant with summary-level Spearman  $\rho$  averaged over the four dimensions, and select the best  $\lambda^*$ . The selected  $\lambda^*$  is then re-evaluated on the test

Table 3: Robustness of the  $\lambda^*$  selection protocol to the choice of dev/test split. The manuscript uses the JSONL-emission-order split (first 50 documents = dev, last 50 = test); here the entire selection procedure — sweep the grid on a dev half, take the argmax, score it on the complementary test half — is re-run on 2000 independent random 50/50 article splits.  $P(\text{win})$  is the fraction of random splits whose dev-argmax lands on that grid value. The bottom row summarises the test-side mean  $\rho$  the protocol would have reported under each random split’s own winner, against the JSONL split’s actual test result.

| $\lambda$                                                                      | $P(\text{win} \mid \text{random split})$ |
|--------------------------------------------------------------------------------|------------------------------------------|
| 0                                                                              | 0.0%                                     |
| 0.25                                                                           | 52.1%                                    |
| 0.3                                                                            | 42.1%                                    |
| 0.4 ★                                                                          | 5.6%                                     |
| 0.5                                                                            | 0.1%                                     |
| 0.6                                                                            | 0.0%                                     |
| 0.75                                                                           | 0.0%                                     |
| 1                                                                              | 0.0%                                     |
| 2                                                                              | 0.0%                                     |
| <hr/>                                                                          |                                          |
| $P(\text{random winner} = \lambda_{\text{JSONL}}^* = .4000): 5.6\%$            |                                          |
| Test $\rho$ at random winner: mean .2900, sd .0198 (JSONL test $\rho$ : .3262) |                                          |
| <hr/>                                                                          |                                          |

split, which is *not* consulted during selection. The verification criterion is whether  $\lambda^*$  outperforms the no-prior baseline ( $\lambda = 0$ ) on the test split’s mean Spearman  $\rho$  — that is, whether the tuned value still helps on data unseen during selection. We do not require the test split’s argmax to match  $\lambda^*$ ; the selection is over the development split only, and a flat optimum (where multiple  $\lambda$  values give near-equal performance) is operationally desirable.

Reporting  $\lambda^*$  as a bare number would leave open how much of it is reproducible, so the selection procedure is itself submitted to the cluster bootstrap of Section 4.5: on each resample of the development articles the full sweep is recomputed and its argmax recorded, giving a distribution over grid values rather than a single winner. Section 7.3 reports that distribution together with two controls that separate a genuine shift in the optimum from ordinary resampling noise. We recommend that any future variant of this protocol report the same distribution; a selection procedure that cannot state how often it would reselect its own output has not finished reporting its result.

#### 4.4. Choice of correlation coefficient

We report Spearman  $\rho$  as the primary correlation coefficient and Kendall  $\tau$  as a secondary check; Pearson  $r$  is computed internally for diagnostics, but is not shown in the headline tables. The choice of Spearman is consistent with prior SUMMEVAL meta-evaluation: human ratings on the four dimensions are ordinal Likert scores, and the metric value is a real number whose absolute calibration we do not claim. A rank-based correlation is the appropriate statistic for “does the metric order summaries the way the rater does”, which is the operational question. Pearson would additionally penalise non-linear monotonic relationships between the metric and the human score that are immaterial for ranking, and is therefore over-restrictive for the use case.

#### 4.5. Bootstrap and significance testing

For each correlation, we report a 95% confidence interval estimated by *cluster bootstrap* at the article level. This requires resampling the documents with replacement and computing the correlation on the concatenated set of sample-level scores in the resampled documents, with 1,000 resamples for the headline tables and 5,000 for the finer-grained sweeps in Section 7.2. The article-level clustering is essential: SUMMEVAL’s 16 candidates per article are not independent observations, since they share a source whose intrinsic difficulty depresses or lifts all 16 scores together. A naïve sample-level bootstrap would treat the 1,600 pairs as independent, underestimate the variance, and produce confidence intervals that are systematically too narrow [42]. The cluster bootstrap is the canonical fix and is what SUMMEVAL-era meta-evaluation uses.

For metric-vs-baseline comparisons we use a *paired* bootstrap over articles (5,000 resamples): on each resample we compute both metrics’ Spearman on the same resample, record the delta  $\Delta\rho = \rho_{\text{LGS}} - \rho_{\text{baseline}}$ , and report the mean delta, its 95% confidence interval, and a two-sided  $p$ -value computed as twice the smaller of the proportions of  $\Delta$  at or below zero versus at or above zero. Pairing is critical: the two metrics’ correlations co-move with the difficulty of the resampled article subset, and an unpaired comparison of two separately-bootstrapped confidence intervals would inflate the apparent uncertainty by exactly that shared component. The paired procedure also yields directly interpretable “LGS-significantly-better-than-baseline” verdicts, which is the form the comparison table in Section 7 reports.

*Multiple comparisons.* The comparison table tests one hypothesis family — “does LGS correlate with human judgment differently from this baseline on this dimension?” — once per (baseline, dimension) cell, for  $6 \times 4 = 24$  cells. Reading 24 raw  $p$ -values against a per-test threshold of 0.05 would put the expected number of spurious positives near one even if no real difference existed, so every cell also carries  $q$ , its Benjamini–Hochberg adjusted  $p$ -value over the whole table [43]. Controlling the false discovery rate rather than the family-wise error rate is the appropriate choice here because the cells are numerous and positively dependent: they share the target metric’s score vector and are computed on the same resampled articles. A cell is reported as a positive finding only when  $q < 0.05$  *and* the paired-bootstrap interval for  $\Delta\rho$  excludes zero. Every significant verdict in Table 9 survives the correction; the weakest survivor is the coherence comparison against SMART-Model at  $q = .010$ .

#### 4.6. Final scoring

After verification, the headline correlations reported in Section 7 are computed on the full 100-article SUMMEVAL set with  $\lambda = \lambda^*$ . The development/test split is used *only* for hyperparameter selection; the headline numbers are not the test-split numbers.

## 5. Experimental Setup

### 5.1. Dataset

SUMMEVAL [9] is built on the CNN/DailyMail reading-comprehension corpus [40]: 100 CNN/DailyMail articles  $\times$  16 abstractive summarisation systems = 1,600 (source, candidate) pairs. Each pair has expert human ratings on coherence (Coh), consistency (Con), fluency (Flu) and relevance (Rel) on a 1–5 scale. We use summary-level correlations as the primary measure (Spearman  $\rho$ , Kendall  $\tau$ ); system-level correlations are reported as a secondary check.

### 5.2. Embedding backbones

The canonical run of LGS uses `nomic-embed-text` (137M parameters) [39], served locally via Ollama, and fixed before the backbone ablation was run. We additionally repeat the canonical metric in Section 7.4 with `all-minilm` (23M, derived from [44]), `mxbai-embed-large` (335M), and `bge-m3` (567M) [41] to verify that the metric design is not specific to one backbone.

### 5.3. Baselines and the controlled-budget protocol

We compare against thirteen baselines spanning the reference-based, reference-free, and mixed families. The *reference-based* family contains BLEU [1], ROUGE-L [2], ChrF [4], METEOR [3], SMART-String, SMART-Model (both [8]), EMBEDSCORER (our whole-text cosine baseline on the same embedder as LGS, included as the like-for-like control), BERTScore [5], MoverScore [6], BARTScore [7], and GPTScore [29]. The *source-based* family contains G-Eval [11]. The *mixed* family — metrics that consume both the source article and a single reference summary — contains UniEval [10].

Because the object of study is the cost–quality trade-off, all fourteen metrics are run under one protocol: the same machine, the same single-threaded runtime harness, the same 1,600 pairs, and the same wall-clock instrumentation. Enforcing a single budget has a consequence that must be stated plainly: the model-based baselines run in configurations smaller than those of the originating papers. Table 4 lists every deviation and the expected direction of its effect on that baseline’s correlation. This is a deliberate trade — a like-for-like cost axis in exchange for absolute correlation heights that are not directly comparable with published figures — and it is the reason the comparisons in this paper are internal to our reproduction.

These deviations shift the absolute correlation heights of the affected baselines slightly downward, but they do not invert the ordering of metric families on SUMMEVAL: UniEval and G-Eval remain the strongest reference-free contenders even under the budget configurations, and the LLM-judge metric stays distinguishable from the embedding-only metrics by mean correlation. All claims concerning LGS are made against *these reproductions*, not against the original-paper numbers; we state this explicitly so that a reader comparing tables across papers can follow the argument. Minor differences between values reported in the original papers and our computed values are expected in any case, given different hardware and the nondeterminism present in several of the methods.

### 5.4. Significance testing

For each (LGS, baseline, dimension) cell we run a paired bootstrap over articles (5,000 resamples) on summary-level Spearman, and report the mean delta, the 95% confidence interval, and a two-sided  $p$ -value. The full procedure is described in Section 4.5.

The paired-bootstrap table above tests LGS against six chosen baselines one dimension at a time. To ask the question over the *whole* pool at once

Table 4: Per-baseline deviations from the originally published configurations, imposed by the single-machine budget of the controlled protocol. “Impact” summarises the expected direction on the absolute correlation height of the affected baseline; the qualitative ordering of metric *families* on SUMMEVAL is unchanged in every case.

| Metric     | Original configuration                                                           | Our reproduction                                                                                                                        | Impact                                                         |
|------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| MoverScore | Corpus-specific IDF weights over tokens; ELMo / BERT contextual embeddings.      | Uniform IDF; otherwise unchanged.                                                                                                       | Slight ↓ (isolates transport cost from IDF re-weighting).      |
| GPTScore   | GPT-3 (175B) as the conditional log-likelihood scorer.                           | GPT-2 (~124M) as the conditional log-likelihood scorer.                                                                                 | Moderate ↓ (much smaller LM, weaker calibration).              |
| G-Eval     | GPT-4, chain-of-thought rubric, $N=20$ sampled judgments per pair.               | qwen2.5:7b-instruct-q4_K_M [45] via Ollama [46], $N=3$ runs at $T=0.7$ , across-run mean used as canonical score and mean±std reported. | Slight-to-moderate ↓ (smaller judge LLM; budget reproduction). |
| UniEval    | Fine-tuned T5-large per dimension, Boolean query, optional verifier-model chain. | Single Boolean query per (article, candidate, dimension); released T5-large checkpoint, no verifier chain.                              | Slight ↓ (drops the verifier-model refinement).                |

we additionally run a Friedman test (Table 5), blocked by article rather than by dimension: with only four SummEval dimensions, using them as blocks gives four data points and no statistical power, so instead each of the 100 articles is one block and a metric’s response in that block is its own Spearman correlation with human ratings computed over the 16 systems of that article. This gives 100 blocks  $\times$  17 treatments (the fourteen published metrics plus the three lead-window controls of Section 6.3) per dimension — enough to reject the null of equal average rank in every dimension ( $\chi_F^2$ ,  $p < .001$  throughout). Post-hoc, Wilcoxon signed-rank tests of LGS against every other metric, Holm-corrected within each dimension’s family of comparisons, confirm the pattern already visible in the paired-bootstrap table and the confound analysis of Section 6.4: LGS is statistically indistinguishable from BERTScore and G-Eval on several dimensions, significantly behind UniEval on every dimension, and — the result that matters most for the Pareto claim above — not significantly different from the lead-window baselines on

coherence in one direction and behind them on consistency and fluency in the other, rather than uniformly ahead as a headline "LGS beats the trivial baseline" framing would suggest.

Table 5: Friedman test (blocks = 100 SUMMEVAL articles, treatments = metrics, response = each metric’s within-article Spearman correlation with human ratings) and Wilcoxon signed-rank post-hoc comparisons of LGS against every other metric, Holm-corrected across the family of comparisons within each dimension.  $\chi_F^2$  is the Friedman statistic (df in parentheses); the omnibus test rejects the null of equal average rank across all four dimensions. The Nemenyi critical difference (CD,  $\alpha = 0.05$ ) is the minimum gap in average rank for two metrics to be called significantly different without a paired test. Wilcoxon columns report LGS’s average-rank gap against the listed baseline and the Holm-adjusted  $p$ -value; **bold** survives  $p_{\text{Holm}} < 0.05$ .

*Coherence:*  $\chi_F^2(16)=346.3$ ,  $p < .001$ ,  $CD=2.47$

| Metric         | Mean $\rho$ | Avg. rank | $z$ (vs LGS) | $p_{\text{Holm}}$ |
|----------------|-------------|-----------|--------------|-------------------|
| UniEval        | 0.552       | 14.38     | -5.17        | <b>0.000</b>      |
| BERTScore      | 0.358       | 11.68     | -0.09        | 1.000             |
| <b>LGS</b>     | 0.370       | 11.50     | —            | —                 |
| G-Eval         | 0.379       | 11.14     | +0.08        | 1.000             |
| GPTScore       | 0.355       | 10.90     | +0.97        | 0.994             |
| SMART-Model    | 0.279       | 9.79      | +3.01        | <b>0.010</b>      |
| EmbedScorer    | 0.239       | 9.21      | +3.74        | <b>0.001</b>      |
| Lead-5 (sent)  | 0.232       | 8.54      | +5.33        | <b>0.000</b>      |
| Lead-3 (sent)  | 0.228       | 8.35      | +6.11        | <b>0.000</b>      |
| Lead-3 (whole) | 0.228       | 8.27      | +5.61        | <b>0.000</b>      |
| ROUGE-L        | 0.179       | 8.20      | +4.17        | <b>0.000</b>      |
| BARTScore      | 0.206       | 8.08      | +4.39        | <b>0.000</b>      |
| METEOR         | 0.163       | 7.95      | +4.33        | <b>0.000</b>      |
| ChrF           | 0.158       | 7.17      | +5.29        | <b>0.000</b>      |
| MoverScore     | 0.090       | 6.60      | +5.76        | <b>0.000</b>      |
| BLEU           | 0.081       | 6.51      | +5.84        | <b>0.000</b>      |
| SMART-String   | -0.014      | 4.75      | +6.95        | <b>0.000</b>      |

*Consistency:*  $\chi_F^2(16)=370.0$ ,  $p < .001$ ,  $CD=2.47$

| Metric         | Mean $\rho$ | Avg. rank | $z$ (vs LGS) | $p_{\text{Holm}}$ |
|----------------|-------------|-----------|--------------|-------------------|
| G-Eval         | 0.492       | 14.49     | -6.78        | <b>0.000</b>      |
| Lead-5 (sent)  | 0.361       | 12.29     | -3.84        | <b>0.001</b>      |
| UniEval        | 0.369       | 11.73     | -3.47        | <b>0.003</b>      |
| Lead-3 (sent)  | 0.330       | 11.31     | -2.28        | 0.067             |
| Lead-3 (whole) | 0.310       | 10.51     | -1.31        | 0.383             |
| <b>LGS</b>     | 0.267       | 10.02     | —            | —                 |
| GPTScore       | 0.239       | 9.55      | +0.88        | 0.383             |
| ChrF           | 0.189       | 8.54      | +2.42        | 0.062             |
| BARTScore      | 0.172       | 23 8.02   | +3.05        | <b>0.011</b>      |
| ROUGE-L        | 0.144       | 7.92      | +3.72        | <b>0.001</b>      |
| BLEU           | 0.138       | 7.79      | +3.91        | <b>0.001</b>      |
| SMART-String   | 0.112       | 7.41      | +4.33        | <b>0.000</b>      |
| SMART-Model    | 0.095       | 7.35      | +5.56        | <b>0.000</b>      |
| METEOR         | 0.095       | 7.29      | +5.20        | <b>0.000</b>      |
| BERTScore      | 0.099       | 7.21      | +4.73        | <b>0.000</b>      |
| EmbedScore     | 0.049       | 6.59      | +5.77        | <b>0.000</b>      |

## 6. Cost–Quality Analysis

Metric papers report how well a metric agrees with human raters. They do not report what that agreement costs. For a metric that will be called inside an automated decision loop, the second number is as binding as the first, and without it one cannot detect the situation that turns out to be common in the current pool: a metric that is both slower and weaker than an alternative already on the shelf. This section supplies the missing axis.

### 6.1. Per-sample cost

The deployment cost of LGS is one sentence-embedder forward pass per source sentence and per candidate sentence. With `nomic-embed-text` (137M parameters) served via Ollama on a single Apple-silicon CPU thread and per-document caching of source embeddings, the canonical SUMMEVAL run — 1,600 (source, candidate) pairs spanning approximately  $6.3 \times 10^3$  cached sentence embeddings (1,273 source plus 5,002 candidate) — completes in roughly ninety seconds ( $\sim 57$  ms per sample; Figure 2). By contrast, G-Eval at our reproduction scale (`qwen2.5:7b-instruct-q4_K_M`,  $N=3$  samples per candidate at temperature 0.7, four dimensions evaluated independently) requires approximately 2.1 hours per single-run pass and approximately 6 hours per  $N=3$  multi-run pass on the same hardware. UniEval requires one forward pass of a fine-tuned T5-large [28] per (article, candidate, dimension) tuple, four dimensions  $\times$  1,600 candidates, totalling approximately 12 minutes wall-clock in our reproduction ( $\sim 430$  ms per sample); the pretrained checkpoints are served via the auxiliary Python server in `cmd/modelsrv`, built on top of the HuggingFace `transformers` library [47]. For metrics that score one dimension at a time — G-Eval and UniEval — the per-sample cost reported throughout is the sum over the four dimensions, because a deployed dimensional scorer must invoke the model once per dimension.

### 6.2. The cost–quality plane

Figure 2 plots all fourteen metrics on a single cost–quality plane: the x-axis is the per-sample wall-clock runtime in milliseconds (log scale), and the y-axis is the mean summary-level Spearman  $\rho$  across the four SUMMEVAL dimensions. Three regions are immediately readable. The bottom-left band contains the n-gram metrics (BLEU, ROUGE-L, METEOR, ChrF, SMART-String): sub-millisecond cost, mean  $\rho$  in the  $[.08, .19]$  range. The middle band contains the contextual-embedding reference-based metrics

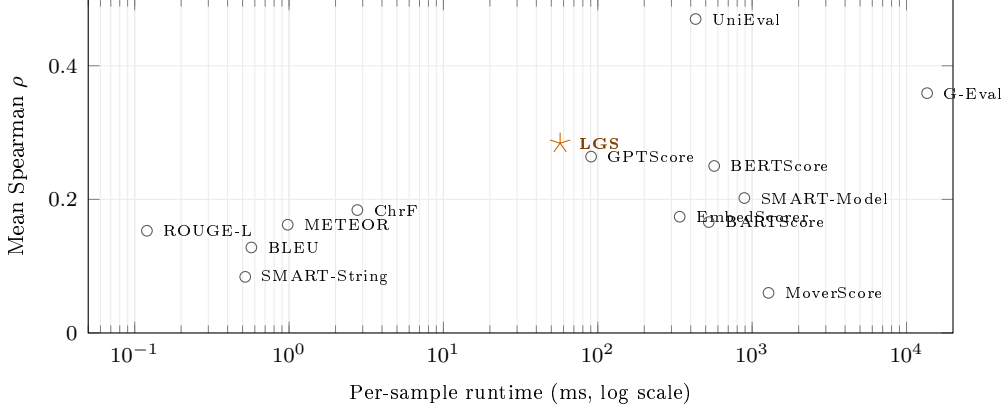


Figure 2: Cost-accuracy frontier on SUMMEVAL. Each point is one metric: x-axis is the per-sample wall-clock runtime under our reproduction (Section 6.1), in milliseconds on a single Apple-silicon CPU thread, on a log scale; y-axis is the mean summary-level Spearman  $\rho$  across the four SUMMEVAL dimensions. LGS (orange star) sits in the lower-cost band of the upper-correlation cluster, distinct from the LLM-judge and fine-tuned-encoder metrics in the upper-right.

(BERTScore, MoverScore, BARTScore, EmbedScorer, SMART-Model): hundreds of milliseconds to a couple of seconds per sample, mean  $\rho$  in [.06, .25]. The upper band contains the LLM-judge and fine-tuned-encoder metrics (G-Eval, UniEval): hundreds of milliseconds to ten-plus seconds per sample, mean  $\rho$  in [.36, .47]. LGS lands in a distinct location: a per-sample cost 6 to 22 times below the contextual-embedding band ( $\sim 57$  ms, dominated by the sentence embedder rather than tokenisation), and a mean  $\rho$  ( $\sim .28$ ) above every metric in that band.

### 6.3. Pareto analysis

The plane admits a sharper reading than visual inspection. Call a metric *dominated* if some other metric in the pool is at least as fast and at least as well correlated, with at least one of the two strict; the metrics that are not dominated form the Pareto frontier of the pool. A reviewer of an earlier version of this paper asked the obvious control question that the fourteen-metric pool above does not answer: is any of this beating the trivial thing? We therefore extend the pool with three reference-free lead-window controls (`cmd/leadbaseline`): keep the first  $k$  source sentences and score the candidate against them with ROUGE-L, either as one block (“whole”) or, reusing LGS’s own mean-of-max aggregation, sentence by sentence against the  $k$  re-

tained source sentences (“sent”). Both variants use no model, no reference summary and no hyperparameter beyond the integer  $k$ . Table 6 reports the outcome over all seventeen metrics.

**The lead-window controls dominate LGS.** Lead-5 (sent) costs 0.06 ms per sample — roughly  $950\times$  cheaper than LGS’s 57 ms — and attains a higher raw mean Spearman  $\rho$  (.296 vs .284). Lead-3 (sent) costs 0.04 ms and ties LGS at  $\rho = .284$ . Once these three lines of ROUGE-L matching are in the pool, only four of the seventeen metrics are Pareto-optimal: Lead-3 (whole), Lead-3 (sent), Lead-5 (sent) and UniEval. LGS is strictly dominated: the same sentence-level mean-of-max aggregation LGS uses, run with a positional cutoff and lexical overlap instead of an embedder and an exponential prior, is both cheaper and better correlated on the raw axis this paper’s Pareto argument uses. This is the most damaging result in this paper and we report it plainly rather than reframing the pool to hide it.

*Why the trivial control wins: an extractiveness confound.* Section 6.4 shows why: a lead-window control does not measure summary quality, it measures how much of the source the candidate copies verbatim, and SUMMEVAL’s human ratings reward that copying. Once this confound is controlled for, LGS recovers relative to the lead controls (Table 7); the raw-axis domination above is real, but it is a statement about what the raw axis of this benchmark measures, not a clean loss for LGS’s grounding mechanism.

*Among learned and embedding-based metrics, LGS still dominates six.* Restricted to the thirteen published, non-trivial baselines, LGS is simultaneously faster and better correlated on the raw axis than EMBEDSCORER ( $6.0\times$ ,  $+.110$ ), GPSTScore ( $1.6\times$ ,  $+.020$ ), BARTScore ( $9.2\times$ ,  $+.118$ ), BERTScore ( $10.0\times$ ,  $+.034$ ), SMART-Model ( $15.6\times$ ,  $+.082$ ) and MoverScore ( $22.4\times$ ,  $+.224$ ), where the first figure is the runtime ratio and the second the difference in mean summary-level Spearman  $\rho$ . The BERTScore comparison is the notable one on the raw axis: a 137M-parameter sentence embedder with no access to the human reference summary attains a higher mean correlation than a RoBERTa-large token-matching metric that does have that access, at one tenth of the runtime; on the confound-controlled axis the two are statistically tied (Section 6.4). We are careful about the strength of the raw-axis statement — at the level of individual dimensions the paired bootstrap confirms a significant advantage only on consistency (Section 7) — but among this restricted pool the domination in the (cost, mean  $\rho$ )

Table 6: Cost-quality Pareto analysis of the fourteen published metrics plus three reference-free lead-window controls, ordered by per-sample runtime. Cost is wall-clock milliseconds per (source, candidate) pair under the controlled protocol of Section 5, summed over the four dimensions for dimensional scorers;  $\bar{\rho}$  is the mean summary-level Spearman across the four SUMMEVAL dimensions on the raw (uncontrolled) axis. A metric is dominated when another is no slower and no less correlated. Values are those of Figure 2, Table 8 and Table 7.

| Metric               | Input         | ms / sample | $\bar{\rho}$ | Status                     |
|----------------------|---------------|-------------|--------------|----------------------------|
| Lead-3 (whole block) | source        | 0.02        | .267         | Pareto-optimal             |
| Lead-3 (sent)        | source        | 0.04        | .284         | Pareto-optimal             |
| Lead-5 (sent)        | source        | 0.06        | .296         | Pareto-optimal             |
| ROUGE-L              | reference     | 0.12        | .153         | dominated by Lead-5 (sent) |
| SMART-String         | reference     | 0.52        | .084         | dominated by Lead-5 (sent) |
| BLEU                 | reference     | 0.57        | .128         | dominated by Lead-5 (sent) |
| METEOR               | reference     | 0.98        | .162         | dominated by Lead-5 (sent) |
| ChrF                 | reference     | 2.77        | .184         | dominated by Lead-5 (sent) |
| <b>LGS (ours)</b>    | source        | 57.01       | .284         | dominated by Lead-5 (sent) |
| GPTScore             | reference     | 90.53       | .264         | dominated by LGS           |
| EMBEDSCORER          | reference     | 339.33      | .174         | dominated by LGS           |
| UniEval              | source + ref. | 430.31      | .470         | Pareto-optimal             |
| BARTScore            | reference     | 523.46      | .166         | dominated by LGS           |
| BERTScore            | reference     | 567.66      | .250         | dominated by LGS           |
| SMART-Model          | reference     | 890.02      | .202         | dominated by LGS           |
| MoverScore           | reference     | 1276.58     | .060         | dominated by LGS           |
| G-Eval               | source        | 13593.99    | .359         | dominated by UniEval       |

plane is unambiguous. It is the pool a practitioner reaches for after already deciding a model-based metric is worth the engineering overhead; it is not the pool a reviewer should accept as establishing Pareto-optimality without qualification.

*The LLM judge is dominated in our reproduction.* G-Eval costs  $31.6\times$  more per sample than UniEval and attains a lower mean  $\rho$  (.359 vs .470), so it is off the frontier entirely. This is a statement about the budget configuration of Table 4, not about G-Eval with a frontier-grade judge; it is nonetheless the relevant statement for a practitioner who must run evaluation on their own hardware, and it illustrates why the cost axis changes conclusions that a correlation-only table would support.

Table 7: Extractiveness confound on SUMMEVAL. The copy rate of a candidate is the fraction of its token 2-grams that occur in the source; it requires no model, no reference and no hyperparameter.  $\bar{\rho}$  is the mean summary-level Spearman correlation with human ratings across the four dimensions,  $\bar{\rho}_{\text{part}}$  the same quantity with the copy rate partialled out of both sides, and  $\rho_{\text{copy}}$  the correlation between the metric and the copy rate. Brackets give the 95% cluster-bootstrap interval over articles. The final row is the copy rate scored as if it were a metric: on the raw axis it outranks most of the pool. Pareto status is given on both axes.

| Metric                | ms/sample | $\bar{\rho}$ | $\bar{\rho}_{\text{part}}$ | $\rho_{\text{copy}}$ | Pareto            |
|-----------------------|-----------|--------------|----------------------------|----------------------|-------------------|
| UniEval               | 430.31    | .470         | .424 [.397, .451]          | .396                 | raw, partial      |
| G-Eval                | 13593.99  | .359         | .318 [.286, .349]          | .404                 | —                 |
| BERTScore             | 567.66    | .250         | .241 [.194, .283]          | .058                 | —                 |
| <b>LGS</b>            | 57.01     | .284         | .198 [.158, .238]          | .386                 | partial           |
| SMART-Model           | 890.02    | .202         | .182 [.140, .225]          | .096                 | —                 |
| GPTScore              | 90.53     | .264         | .165 [.124, .205]          | .433                 | —                 |
| METEOR                | 0.98      | .162         | .165 [.123, .205]          | .013                 | partial           |
| EmbedScorer           | 339.33    | .174         | .164 [.121, .207]          | .052                 | —                 |
| Lead-5 (mean-of-max)  | 0.06      | .296         | .154 [.115, .192]          | .684                 | raw, partial      |
| Lead-3 (mean-of-max)  | 0.04      | .284         | .154 [.113, .195]          | .607                 | raw, partial      |
| Lead-3 (whole block)  | 0.02      | .267         | .144 [.104, .185]          | .550                 | raw, partial      |
| BARTScore             | 523.46    | .166         | .136 [.092, .181]          | .128                 | —                 |
| ChrF                  | 2.77      | .184         | .128 [.083, .170]          | .232                 | —                 |
| ROUGE-L               | 0.12      | .153         | .127 [.077, .172]          | .115                 | —                 |
| BLEU                  | 0.57      | .128         | .091 [.044, .136]          | .149                 | —                 |
| MoverScore            | 1276.58   | .060         | .091 [.045, .137]          | -.102                | —                 |
| SMART-String          | 0.52      | .084         | .081 [.036, .125]          | .025                 | —                 |
| Copy rate (no metric) | 0.00      | .275         | .000                       | 1.000                | <i>diagnostic</i> |

#### 6.4. The extractiveness confound

The lead-window result above raises an uncomfortable question: is a four-line ROUGE-L heuristic really capturing summary *quality*, or is it capturing something else that SUMMEVAL’s human ratings happen to reward? We test the latter directly. Define the *copy rate* of a candidate as the fraction of its token bigrams that occur verbatim in the source — no model, no reference, no hyperparameter. Table 7 reports, for every metric in the pool, its raw correlation  $\bar{\rho}$ , its correlation with the copy rate itself ( $\rho_{\text{copy}}$ ), and its correlation with human judgment after the copy rate is partialled out of both sides ( $\bar{\rho}_{\text{part}}$ ).

Scored as if it were a metric, copy rate alone attains  $\bar{\rho} = .275$ , which would outrank eleven of the fourteen published metrics in our pool, including BERTScore, GPTScore and every lexical metric. The lead-window controls are, unsurprisingly, close to pure copy detectors ( $\rho_{\text{copy}} = .607\text{--}.684$ ); LGS is also substantially confounded ( $\rho_{\text{copy}} = .386$ ), more so than BERTScore (.058), which never observes the source and so cannot leak extractiveness into its score by construction. On the confound-controlled axis LGS’s point estimate rises relative to the lead controls ( $\bar{\rho}_{\text{part}} = .198$  vs .154 for both Lead-3 and Lead-5), but a paired bootstrap on that axis (5,000 resamples) does not support a confident separation: LGS vs Lead-5,  $\Delta = +.044$ , 95% CI  $[+.004, +.084]$ ,  $p = .033$  (marginal); LGS vs Lead-3,  $\Delta = +.045$ , 95% CI  $[-.003, +.090]$ ,  $p = .064$  (not significant); LGS vs BERTScore,  $\Delta = -.043$ , 95% CI  $[-.090, +.004]$ ,  $p = .074$  (a tie, numerically favouring BERTScore).

We read this as follows. The raw-axis Pareto domination of Section 6.3 is real but is substantially a statement about what SUMMEVAL’s mean-correlation axis measures, not a clean verdict on LGS’s sentence-level grounding mechanism. Controlling for the confound, LGS cannot be defended as contributing something a lexical lead heuristic does not: it is not statistically distinguishable from Lead-3. It also does not separate from BERTScore, a metric with access to the human reference that LGS does not use. We regard this as the central limitation of this paper’s central metric, and we discuss its implications for the field’s evaluation practice, not only for LGS, in Section 8.4.

#### 6.5. *What the analysis shows and does not show*

The Pareto reading, once the lead-window controls and the confound-controlled axis are both in view, supports a narrower thesis than the raw table alone would suggest: among learned and embedding-based metrics LGS occupies an undominated cost–quality point, but it does not by itself rank the metrics on quality, and it does not survive as Pareto-optimal against trivial reference-free baselines on the axis this benchmark reports by default. The y-axis is mean Spearman, which hides per-dimension variation (Section 7.4), and the x-axis is wall-clock runtime under our specific reproduction stack, not theoretical asymptotic complexity. Four caveats should be kept in mind. First, the LLM-judge cost depends linearly on the number of judge calls per pair, which we set at  $N=3$ ; a single-run G-Eval at  $N=1$  would shift left by a factor of three but would remain dominated by UniEval on both axes.

Second, the embedding-metric costs depend on whether source-side embeddings are cached across candidates, which we do in LGS; without caching the per-pair cost would multiply by the candidate-to-source-sentence ratio, although in absolute terms it would still remain in the embedding band rather than crossing into the LLM-judge band. Third, runtimes reflect a CPU-only single-thread deployment target; GPU serving would compress the differences between model-based metrics, though not the difference between one small embedder pass and  $N$  autoregressive decodes. Fourth, the mean over four dimensions is a summary statistic chosen for readability; a practitioner optimising a single dimension should read Table 8 directly. We discuss the deployment scenarios that put the frontier point to operational use in Section 8.1.

## 7. Results

Table 8 reports summary-level Spearman  $\rho$  correlations with human judgment for all fourteen metrics. Table 9 reports the paired-bootstrap deltas of LGS against each baseline on Spearman  $\rho$ . Table 10 reports system-level correlations (note that with  $n = 16$  systems the confidence intervals are wide; we treat this table as a secondary sanity check). The corresponding Kendall  $\tau$  tables are reported in Appendix Appendix A. The two coefficients induce almost the same ordering of the fourteen metrics: the rank correlation between the  $\rho$ -ordering and the  $\tau$ -ordering is .987–1.000 across the four dimensions, and no metric changes rank by more than two positions. The discussion below is therefore phrased in terms of  $\rho$  alone (Section 4.4).

### 7.1. Headline findings

*Ablation of the knowledge component.* The controlled test of the LGS design is the comparison against EMBEDSCORER, the whole-text cosine baseline on the same `nomic-embed-text` backbone (Section 2). LGS significantly outperforms EMBEDSCORER on coherence, consistency, and fluency ( $\Delta\rho = +.117/+ .171/+ .116$ , all  $p < .001$  by paired bootstrap with 5,000 resamples) and is in a statistical tie on relevance. Because the only difference between the two metrics is architectural — sentence-level grounding via  $\max_i$  on the source plus the lead-bias prior  $w(i)$ , versus a single whole-text cosine — the lift is directly attributable to those choices and cannot be explained by embedder strength. The prior itself is ablated separately in Section 7.2

Table 8: Summary-level correlations with human judgment on SummEval.  $\rho$  = Spearman. Brackets give the 95% cluster-bootstrap confidence interval over articles.

| Metric       | Coh                 | Con                 | Flu                | Rel               |
|--------------|---------------------|---------------------|--------------------|-------------------|
| BLEU         | .102 [.036, .165]   | .137 [.085, .185]   | .066 [.002, .123]  | .206 [.123, .277] |
| ROUGE-L      | .163 [.093, .231]   | .140 [.085, .195]   | .108 [.043, .167]  | .201 [.115, .278] |
| ChrF         | .158 [.092, .219]   | .174 [.125, .220]   | .110 [.050, .167]  | .293 [.219, .361] |
| METEOR       | .199 [.146, .250]   | .134 [.085, .179]   | .130 [.066, .193]  | .185 [.115, .251] |
| SMART-String | -.006 [-.070, .063] | .111 [.054, .166]   | .105 [.046, .159]  | .124 [.045, .194] |
| EmbedScorer  | .279 [.212, .339]   | .056 [-.001, .111]  | .041 [-.025, .107] | .320 [.245, .390] |
| BERTScore    | .377 [.317, .431]   | .109 [.050, .164]   | .142 [.077, .202]  | .371 [.301, .429] |
| MoverScore   | .152 [.085, .218]   | -.026 [-.084, .030] | .011 [-.054, .077] | .102 [.028, .179] |
| SMART-Model  | .328 [.267, .385]   | .093 [.035, .147]   | .067 [.000, .134]  | .321 [.248, .385] |
| BARTScore    | .167 [.102, .229]   | .117 [.055, .176]   | .106 [.037, .167]  | .273 [.192, .348] |
| GPTScore     | .358 [.308, .408]   | .210 [.142, .270]   | .181 [.109, .245]  | .306 [.247, .365] |
| UniEval      | .569 [.529, .606]   | .405 [.367, .443]   | .444 [.398, .485]  | .462 [.412, .509] |
| G-Eval       | .393 [.337, .444]   | .488 [.446, .531]   | .140 [.101, .177]  | .416 [.356, .473] |
| LGS          | .396 [.344, .449]   | .227 [.176, .275]   | .156 [.091, .220]  | .356 [.297, .407] |

by setting  $\lambda=0$ , which isolates the contribution of the injected structural knowledge from that of sentence-level grounding.

*Cross-family comparisons.* Against the broader baseline pool, LGS significantly outperforms SMART-Model on coherence, consistency, and fluency ( $\Delta\rho=+.068/+ .134/+ .089$ ,  $p \leq .005$ ), BERTScore on consistency ( $\Delta\rho=+.118$ ,  $p < .001$ ), and ChrF on coherence ( $\Delta\rho=+.237$ ,  $p < .001$ ). On the remaining dimensions of these comparisons LGS is in a statistical tie, and it exceeds BLEU, ROUGE-L, METEOR, SMART-String, MoverScore and BARTScore on the point estimate of every dimension. Against the top-tier reference-free metrics, LGS loses: it is below UniEval on every dimension by paired bootstrap and below G-Eval specifically on consistency, while tying with G-Eval on coherence, fluency and relevance. We report this as the ceiling of an embedding-only metric of this size. The deployment claim of LGS is the Pareto position of Section 6.3, not the absolute correlation height.

*System-level correlations.* Table 10 reports system-level correlations for all metrics. With  $n=16$  systems the confidence intervals are wide and the per-cell rankings are unstable; we treat the table as a secondary sanity check,

Table 9: Paired bootstrap comparison of LGS against baselines on summary-level Spearman correlations. Each cell reports  $\Delta\rho$  (ours minus baseline) with 95% CI in brackets, the raw two-sided  $p$ -value, and  $q$ , the Benjamini-Hochberg adjusted  $p$ -value over all 24 (baseline, dimension) cells of this table. Bold marks improvements that survive the correction ( $q < 0.05$  and CI excludes 0).

| Baseline    | Coh            | Con            | Flu            | Rel            |
|-------------|----------------|----------------|----------------|----------------|
| UniEval     | <b>-.173</b>   | <b>-.177</b>   | <b>-.287</b>   | <b>-.107</b>   |
|             | [-.236, -.113] | [-.233, -.123] | [-.355, -.226] | [-.176, -.039] |
|             | $p < .001$     | $p < .001$     | $p < .001$     | $p = 0.003$    |
|             | $q < .001$     | $q < .001$     | $q < .001$     | $q = 0.006$    |
| BERTScore   | <b>+.019</b>   | <b>+.118</b>   | <b>+.015</b>   | <b>-.016</b>   |
|             | [-.039, +.076] | [+.059, +.181] | [-.039, +.068] | [-.081, +.052] |
|             | $p = 0.509$    | $p < .001$     | $p = 0.612$    | $p = 0.634$    |
|             | $q = 0.611$    | $q < .001$     | $q = 0.666$    | $q = 0.666$    |
| G-Eval      | <b>+.003</b>   | <b>-.262</b>   | <b>+.017</b>   | <b>-.061</b>   |
|             | [-.076, +.080] | [-.320, -.203] | [-.055, +.086] | [-.144, +.023] |
|             | $p = 0.938$    | $p < .001$     | $p = 0.638$    | $p = 0.152$    |
|             | $q = 0.938$    | $q < .001$     | $q = 0.666$    | $q = 0.243$    |
| SMART-Model | <b>+.068</b>   | <b>+.134</b>   | <b>+.089</b>   | <b>+.035</b>   |
|             | [+.020, +.118] | [+.084, +.182] | [+.038, +.141] | [-.022, +.095] |
|             | $p = 0.005$    | $p < .001$     | $p < .001$     | $p = 0.248$    |
|             | $q = 0.010$    | $q < .001$     | $q < .001$     | $q = 0.331$    |
| ChrF        | <b>+.237</b>   | <b>+.053</b>   | <b>+.046</b>   | <b>+.062</b>   |
|             | [+.145, +.326] | [-.013, +.121] | [-.017, +.113] | [-.031, +.153] |
|             | $p < .001$     | $p = 0.123$    | $p = 0.171$    | $p = 0.188$    |
|             | $q < .001$     | $q = 0.211$    | $q = 0.256$    | $q = 0.265$    |
| EmbedScorer | <b>+.117</b>   | <b>+.171</b>   | <b>+.116</b>   | <b>+.036</b>   |
|             | [+.061, +.174] | [+.122, +.222] | [+.058, +.173] | [-.026, +.101] |
|             | $p < .001$     | $p < .001$     | $p < .001$     | $p = 0.283$    |
|             | $q < .001$     | $q < .001$     | $q < .001$     | $q = 0.357$    |

consistent with prior meta-evaluation practice on SUMMEVAL. The qualitative ordering of the metric families matches the summary-level table.

## 7.2. Ablating the knowledge prior: the exponent $\lambda$

Table 11 reports the development-split sweep of  $\lambda \in \{0, 0.25, 0.5, 1.0, 2.0\}$  and the test-split verification of the selected  $\lambda^* = 0.5$ . Setting  $\lambda=0$  removes

Table 10: System-level correlations with human judgment on SummEval.  $\rho$  = Spearman. Brackets give the 95% cluster-bootstrap confidence interval over articles.

| Metric       | Coh                 | Con                  | Flu                 | Rel                 |
|--------------|---------------------|----------------------|---------------------|---------------------|
| BLEU         | .276 [-.361, .720]  | .024 [-.589, .543]   | .415 [-.223, .818]  | .506 [-.134, .844]  |
| ROUGE-L      | .312 [-.324, .734]  | -.018 [-.607, .526]  | .365 [-.243, .771]  | .482 [-.139, .851]  |
| ChrF         | .526 [.022, .831]   | .794 [.505, .944]    | .832 [.555, .940]   | .803 [.498, .917]   |
| METEOR       | .094 [-.547, .582]  | -.318 [-.728, .229]  | .041 [-.538, .631]  | .179 [-.487, .641]  |
| SMART-String | -.029 [-.777, .511] | -.197 [-.656, .330]  | .171 [-.402, .679]  | .256 [-.468, .735]  |
| EmbedScorer  | .424 [-.097, .790]  | .406 [-.120, .796]   | .262 [-.296, .706]  | .262 [-.243, .745]  |
| BERTScore    | .768 [.406, .955]   | -.029 [-.590, .537]  | .324 [-.294, .818]  | .482 [-.071, .897]  |
| MoverScore   | .109 [-.532, .665]  | -.612 [-.874, -.050] | -.329 [-.836, .277] | -.115 [-.726, .543] |
| SMART-Model  | .700 [.242, .914]   | .594 [.033, .920]    | .538 [-.051, .905]  | .535 [-.056, .920]  |
| BARTScore    | .503 [-.006, .839]  | .812 [.517, .950]    | .785 [.484, .893]   | .776 [.471, .927]   |
| GPTScore     | .612 [.028, .955]   | .538 [-.015, .890]   | .379 [-.281, .858]  | .374 [-.259, .895]  |
| UniEval      | .791 [.407, .970]   | .444 [-.153, .863]   | .762 [.359, .928]   | .815 [.439, .949]   |
| G-Eval       | .526 [.019, .873]   | .902 [.646, .975]    | .259 [-.393, .812]  | .785 [.375, .970]   |
| LGS          | .529 [-.086, .958]  | .709 [.294, .871]    | .488 [-.179, .892]  | .453 [-.127, .943]  |

the injected knowledge exactly, so the difference between the first row and the rest is the measured contribution of the prior. The mean Spearman  $\rho$  across the four dimensions is .233 at  $\lambda = 0$  (no prior) and .252 at  $\lambda^* = 0.5$ , a lift of +.019. On the held-out test split,  $\lambda^* = 0.5$  also outperforms the no-prior baseline on the mean (.319 versus .314). Beyond  $\lambda > 1$  the mean drops, indicating that the prior becomes too aggressive once the article tail is heavily suppressed.

The mean gap between  $\lambda = 0.25$  (.251) and  $\lambda^* = 0.5$  (.252) is .001, well inside the bootstrap noise. To characterise the surface around the optimum we re-ran a finer sweep  $\lambda \in \{0.30, 0.40, 0.50, 0.60, 0.75\}$  with paired bootstrap (5,000 resamples) on both splits. The development surface around the optimum is smooth and flat: per-dimension  $\rho$  values move by less than the bootstrap half-width across the entire  $[0.30, 0.75]$  window, and the test surface mirrors this pattern. On the combined nine-point grid the development argmax is in fact  $\lambda=0.4$  (.2528) rather than the  $\lambda^*=0.5$  (.2519) selected on the five-point grid of the pre-registered protocol; the two differ by .0009. We retain  $\lambda^*=0.5$  because the protocol of Section 4 fixes the grid before the sweep and the finer grid was added afterwards, but the discrepancy is itself evidence that the argmax is not a stable object — which the next section measures

rather than asserts. We therefore read  $\lambda^* = 0.5$  as one representative of a flat plateau rather than an isolated peak.

The per-dimension picture on the test split is more nuanced than on the development split. On development, all four SUMMEVAL dimensions improve at  $\lambda^* = 0.5$  over the no-prior baseline (Coh +.021, Con +.007, Flu +.011, Rel +.036). On test, only the mean and coherence carry that lift cleanly: Coh +.014, Con −.001, Flu +.001, Rel +.003. We therefore claim a meaningful held-out lift for the mean and for coherence; the apparent per-dimension lifts on consistency, fluency, and relevance are within bootstrap noise on the test split and we do not claim them as generalising effects of the lead-bias prior. We also note that the test-split argmax is  $\lambda=0.25$  (mean .334) rather than the development-selected  $\lambda^*=0.5$ ; under the protocol of Section 4 the test split does not participate in selection, and reporting the mismatch is part of the protocol’s value.

Beyond the optimum, the metric degrades gracefully but monotonically. At  $\lambda=1.0$  the development mean drops to .242 and at  $\lambda=2.0$  to .221 — below the no-prior baseline of .233. The mechanism is straightforward to see from the prior’s shape. At  $\lambda=1$ , the weight at the article tail  $w(n-1)$  is  $\exp(-1) \approx 0.37$ ; at  $\lambda=2$  it is  $\exp(-2) \approx 0.14$ . As  $\lambda$  grows, source sentences beyond the lead are increasingly suppressed before the  $\max_i$  operator, and a candidate sentence whose true semantic anchor is in the article body — a paraphrased detail, a quoted figure, a closing context — can no longer reach its anchor with enough weight to win the argmax. The argmax instead collapses onto mismatched lead sentences for these candidates, depressing cos, and  $\rho$  falls. This is the behaviour one should want from an injected prior: it helps when it gently up-weights the lead, and hurts when it functionally truncates the article. The mid-range  $\lambda \in [0.4, 0.6]$  is the operationally useful window, and the calibration sweep is what locates it.

The sweep above selects a single  $\lambda^*$  by the mean Spearman  $\rho$  across all four dimensions, but the optimum is not the same for every dimension. Table 12 makes this explicit: coherence and fluency prefer a mid-range exponent close to the reported aggregate optimum, consistency prefers a weaker prior ( $\hat{\lambda}=0.25$ ), and relevance’s dev-argmax ( $\hat{\lambda}=2$ ) is a selection-noise artefact rather than a robust preference — its test-split  $\rho$  at that value (.336) is *lower* than the no-prior baseline’s test  $\rho$  (.411), the opposite of what a genuine per-dimension optimum should show. A practitioner tuning LGS for a single dimension should consult this table rather than the aggregate  $\lambda^*$ , and should treat the relevance row as a caution about reading any single-

Table 11: Ablation of LGS on SummEval (summary-level Spearman  $\rho$ ). The metric is  $\text{score} = \text{mean}_j \max_i w(i) \cdot \cos(c_j, s_i)$  with exponential lead-bias prior  $w(i) = \exp(-\lambda \cdot i/n)$  on source-sentence position. The exponent  $\lambda$  is selected on a held-out development split (50 articles) and re-evaluated on the test split (50 articles); the dev-selected canonical row is marked  $\star$ .

| Variant                                      | Coh  | Con  | Flu  | Rel  |
|----------------------------------------------|------|------|------|------|
| Recall baseline ( $\lambda = 0$ ) (dev)      | .342 | .180 | .139 | .271 |
| Lead-bias ( $\lambda = 0.25$ ) (dev)         | .358 | .197 | .156 | .294 |
| Lead-bias ( $\lambda = 0.3$ ) (dev)          | .361 | .196 | .156 | .298 |
| Lead-bias ( $\lambda = 0.4$ ) (dev)          | .363 | .191 | .153 | .303 |
| Lead-bias ( $\lambda = 0.5$ ) (dev)          | .363 | .187 | .150 | .307 |
| Lead-bias ( $\lambda = 0.6$ ) (dev)          | .363 | .183 | .145 | .311 |
| Lead-bias ( $\lambda = 0.75$ ) (dev)         | .362 | .175 | .138 | .315 |
| Lead-bias ( $\lambda = 1$ ) (dev)            | .361 | .162 | .126 | .319 |
| Lead-bias ( $\lambda = 2$ ) (dev)            | .349 | .125 | .086 | .322 |
| Recall baseline ( $\lambda = 0$ ) (test)     | .415 | .272 | .161 | .410 |
| Lead-bias ( $\lambda = 0.25$ ) (test)        | .433 | .295 | .180 | .427 |
| Lead-bias ( $\lambda = 0.3$ ) (test)         | .434 | .293 | .178 | .425 |
| Lead-bias ( $\lambda = 0.4$ ) (test)         | .432 | .282 | .171 | .419 |
| Lead-bias ( $\lambda = 0.5$ ) (test) $\star$ | .429 | .271 | .162 | .413 |
| Lead-bias ( $\lambda = 0.6$ ) (test)         | .425 | .261 | .152 | .404 |
| Lead-bias ( $\lambda = 0.75$ ) (test)        | .418 | .248 | .139 | .390 |
| Lead-bias ( $\lambda = 1$ ) (test)           | .402 | .227 | .121 | .369 |
| Lead-bias ( $\lambda = 2$ ) (test)           | .375 | .158 | .079 | .336 |

dimension argmax off a 50-article sweep without its own held-out verification.

Figure 4 decomposes the development-mean curve of Figure 3 (**nomic-embed-text**) into its four SUMMEVAL per-dimension components. The decomposition shows that the  $\lambda^*=0.5$  optimum is a compromise: the four dimensions do not share a single  $\lambda$  that is best for all of them. Coherence and relevance benefit strongly from the prior — coherence peaks broadly in the  $[0.25, 1.0]$  region, relevance continues to climb up to  $\lambda=2$  — while consistency and fluency are best at the *lowest* non-zero  $\lambda$  ( $\lambda=0.25$ ) and deteriorate sharply as  $\lambda$  grows. The mean optimum at  $\lambda=0.5$  is the value that best balances the two pairs; practitioners with a single dimension of interest may prefer a different point on the curve.

Table 12: Per-dimension lead-bias exponent decision matrix. The ablation table selects one  $\lambda^*$  by mean Spearman  $\rho$  across all four dimensions; this table instead selects the dev-argmax  $\lambda$  separately for each dimension and reports it on the held-out test split, alongside the no-prior ( $\lambda=0$ ) reference. A practitioner tuning LGS for a single dimension should read this table, not the aggregate one.

| Dimension   | $\hat{\lambda}$ | Dev $\rho$ | Test $\rho$ | $\lambda=0$ dev | $\lambda=0$ test |
|-------------|-----------------|------------|-------------|-----------------|------------------|
| Coherence   | 0.4             | .3633      | .4321       | .3418           | .4146            |
| Consistency | 0.25            | .1968      | .2950       | .1801           | .2717            |
| Fluency     | 0.3             | .1558      | .1779       | .1388           | .1611            |
| Relevance   | 2               | .3223      | .3363       | .2711           | .4105            |

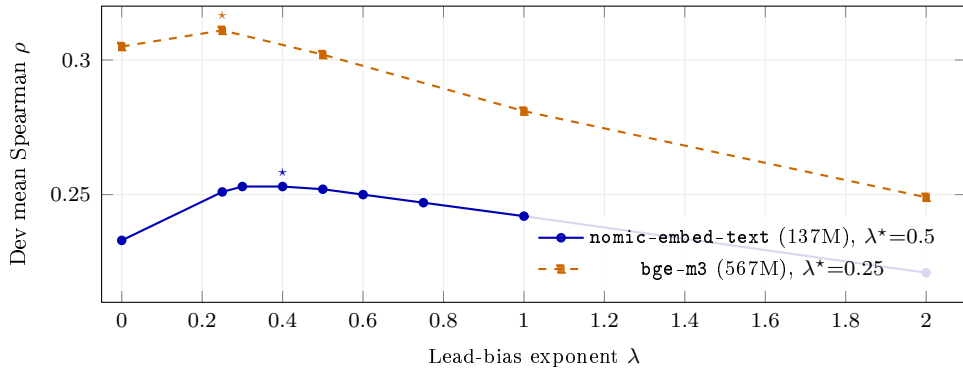


Figure 3: Development-split mean Spearman  $\rho$  across the four SUMMEVAL dimensions as a function of the lead-bias exponent  $\lambda$ , for the two sentence-embedder backbones whose sweep is reported in this paper. Both curves peak at  $\lambda > 0$  and decline monotonically past the peak, but the position of the peak *shifts* between backbones (0.5 for **nomic-embed-text**, 0.25 for **bge-m3**). Reusing the nomic-tuned  $\lambda=0.5$  with **bge-m3** would give  $\rho=0.302$ , below its no-prior baseline of 0.305. This is the empirical motivation for treating the calibration protocol of Section 4 as an obligatory part of the method.

### 7.3. How identifiable is $\lambda^*$ ?

Every number reported so far treats  $\lambda^*$  as if the development split had determined it. It did not determine it; it selected it, and a different draw of 50 articles would have selected something else. Because the entire calibration protocol of Section 4 rests on that selection, we measure how reproducible it is instead of assuming it.

The procedure is the cluster bootstrap of Section 4.5 applied one level up. On each of 5,000 resamples of the development articles we recompute the mean Spearman  $\rho$  at *every* grid value and record which value wins, yielding

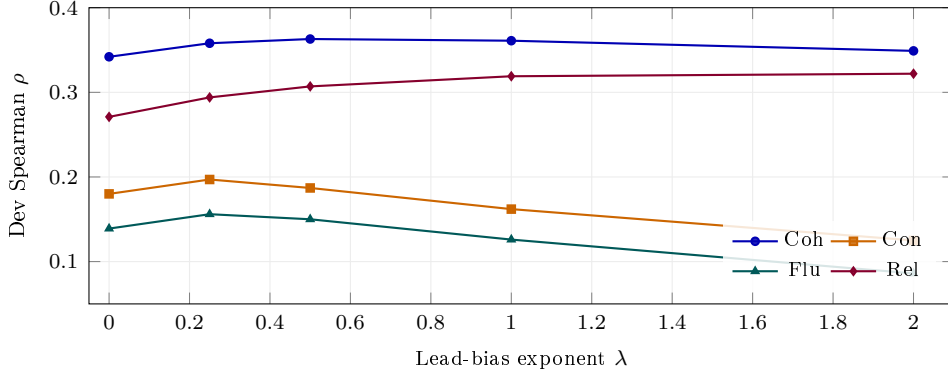


Figure 4: Per-dimension development-split Spearman  $\rho$  versus the lead-bias exponent  $\lambda$  with `nomic-embed-text`. Coherence and relevance benefit from the prior across the swept range ( $\lambda_{\text{Coh}}^*=0.5$ ,  $\lambda_{\text{Rel}}^*=2.0$ ); consistency and fluency peak at the smallest non-zero value ( $\lambda_{\text{Con}}^*, \lambda_{\text{Flu}}^*=0.25$ ) and degrade as  $\lambda$  grows. The canonical  $\lambda^*=0.5$  is the optimum on the mean-of-four criterion (Section 4), not on any single dimension.

a distribution over the grid rather than a point. Table 13 reports it. Because the per-sample scores for each  $\lambda$  are already stored, the measurement requires no additional embedder passes.

*Switching the prior on is reproducible.* The no-prior setting  $\lambda=0$  is selected in 0.7% of development resamples and 0.3% of test resamples. Whatever else is uncertain, the decision to impose *some* positional prior is not: it survives resampling in more than 99% of draws on both splits. This is the claim the ablation supports, and it is weaker than — and logically prior to — any claim about a particular exponent.

*The exponent itself is not identifiable.* The selection mass does not concentrate. On the development split with the canonical backbone,  $\lambda=0.3$  wins 27.6% of resamples,  $\lambda=0.4$  wins 20.4%, the reported  $\lambda^*=0.5$  wins 17.8%, and  $\lambda=0.25$  wins 17.5%: four grid values within ten points of each other, none commanding even a third of the mass. A 50-article development split does not contain enough information to separate them. The practical reading is that the protocol returns the interval  $[0.25, 0.6]$ , inside which no value is distinguishable from any other, and that quoting a three-significant-figure optimum from such a surface overstates what was measured.

*Two controls, and what they do to the backbone claim.* The lower panel of Table 13 compares whole selection outcomes rather than single  $\rho$  values,

on the grid common to every configuration. The control that matters is the third row of the reasoning: a configuration compared *against itself* on two independent draws of the same 50 articles reselects the same exponent only 43.5% of the time, and when it disagrees the direction is a coin flip (28.6% versus 27.9%). That is the noise floor. Against it, the cross-backbone agreement of 37.8% is barely lower — so the observation that two backbones pick different exponents is, on its own, uninformative.

What does separate from the noise floor is the *direction*. When `nomic-embed-text` and `bge-m3` are scored on the same resample and disagree, the former selects the larger exponent in 61.9% of resamples against 0.4% for the reverse, a ratio no symmetric noise process produces. We therefore claim a directional backbone effect — `bge-m3` prefers a weaker prior — and explicitly do not claim a quantified shift from 0.5 to 0.25.

One confound must be stated because it limits even that claim. The same asymmetry appears between the development and test splits under an *unchanged* backbone (52.8% versus 0.9%). Changing which 50 articles are used therefore induces a directional shift of comparable strength to changing the representation space. Our design cannot attribute the cross-backbone asymmetry to the backbone alone, and a clean attribution would need the sweep repeated over many disjoint article samples per backbone — which, on a 100-article benchmark, is not available. Section 8.6 lists the multi-corpus replication that would supply it.

#### 7.4. Backbone ablation

Table 14 shows the canonical  $\lambda=0.5$  metric with four sentence-embedder backbones spanning a  $\sim 24\times$  parameter range: `all-minilm` (23M), `nomic-embed-text` (137M), `mxbai-embed-large` (335M), and `bge-m3` (567M). The mean Spearman across the four SUMMEVAL dimensions falls in  $[\text{.284}, \text{.312}]$  for all four backbones, a span of .028 across the full parameter range. We read this as evidence that the metric *design* — the mean-of-max grounding aggregator with the lead-bias prior — is qualitatively robust to backbone choice: any reasonable sentence embedder produces a working metric.

Two properties of this table need to be stated explicitly, because both cut against the headline numbers rather than for them. First, `nomic-embed-text` — the canonical backbone, fixed before the ablation was run — has the *lowest* mean of the four (.284, against .296 for `mxbai-embed-large`, .298 for `all-minilm` and .312 for `bge-m3`). The headline correlations of Tables 8 and 9, and the Pareto position of Section 6.3, are therefore reported at the

conservative end of the observed band; a backbone swap to **bge-m3** would raise the mean  $\rho$  to .312 without changing the design, at roughly  $5\times$  the embedding cost. We keep the pre-registered canonical backbone rather than the best-performing one, because selecting the backbone on the same data used for the headline table would be a researcher degree of freedom we would rather not exercise.

Second, the per-dimension profile is not stable across backbones. **nomic-embed-text** is by far the strongest on coherence (.396) and relevance (.356) and the weakest on consistency (.227) and fluency (.156); the other three backbones invert this pattern. *Per-dimension* comparisons against baselines — for instance, “LGS exceeds BERTScore on consistency” — are therefore valid for the specific backbone used and should be read as backbone-conditional, not as universal properties of LGS. The mean-Spearman ordering of metric families is stable across backbones; the per-dimension ordering is not.

#### 7.5. Cross-backbone calibration of the prior

The backbone ablation also bears on *where* the prior should be set, not only on how well the metric scores. Figure 3 shows the development-split mean Spearman  $\rho$  as a function of  $\lambda$  for two embedders: **nomic-embed-text** (the canonical backbone) and **bge-m3** (the largest in our roster, chosen as the strongest contrast point). Both curves share their qualitative shape: each peaks at  $\lambda>0$  — that is, the injected structural knowledge helps regardless of the backbone — and declines monotonically past the peak. But the position of the peak *shifts*:  $\lambda^*=0.5$  for **nomic-embed-text** (development mean .252, a lift of +.019 over the no-prior baseline) and  $\lambda^*=0.25$  for **bge-m3** (development mean .311, a lift of only +.006).

The practical consequence is sharp: applying the nomic-tuned  $\lambda=0.5$  to **bge-m3** yields a development mean of .302, which is *below* its no-prior baseline of .305. A prior that helps when its exponent is calibrated against the backbone in use can therefore cost more than it returns when an exponent is carried over from elsewhere.

Two qualifications are essential, and Section 7.3 supplies both. First, the gap between .302 and .305 is a fraction of the bootstrap half-width, so this particular pair of numbers is an illustration rather than a demonstration. What carries statistical weight is the directional result: scored on the same resample, **nomic-embed-text** selects the larger exponent 61.9% of the time against 0.4% for the reverse. Second, that asymmetry is not uniquely

attributable to the backbone, because a change of article sample under a fixed backbone produces one of comparable strength. The defensible claim is therefore that the exponent should be re-derived in situ — not that the backbone is the specific variable that invalidates a borrowed value. That weaker claim is sufficient for the protocol, since the sweep is what transfers and it costs no additional embedder passes.

We hypothesise that the size of the lift depends on the density of the backbone’s representation. `bge-m3` already produces a stronger position-agnostic recall (.305 development mean at  $\lambda=0$ , versus .233 for `nomic-embed-text`), leaving less room for a positional prior to add value. We flag this as a hypothesis rather than a result: with two backbones swept and an exponent that Section 7.3 shows to be poorly identified, the evidence does not discriminate between it and simpler accounts. The general lesson we do draw is weaker and, we think, more durable: an a priori structural prior and a learned representation are not independent components, and how much the explicit prior can still contribute is not knowable without measurement on the deployment’s own data.

Table 13: Selection uncertainty of the lead-bias exponent. For each (split, backbone) the whole selection procedure — compute mean Spearman  $\rho$  across the four SUMMEVAL dimensions at every grid value, take the argmax — is re-run on 5000 cluster-bootstrap resamples over articles.  $P(\arg \max)$  is the fraction of resamples in which that grid value is selected; a selection procedure that identifies a well-separated optimum concentrates this mass on one row.  $\lambda^*$  marks the argmax of the point estimate, i.e. the value the ablation table reports.

| Split / backbone        | $\lambda$ | Mean $\rho$ | 95% CI       | $P(\arg \max)$ |
|-------------------------|-----------|-------------|--------------|----------------|
| dev / bge-m3            | 0         | .305        | [.248, .361] | 13.8%          |
|                         | 0.25 ★    | .311        | [.254, .366] | 84.2%          |
|                         | 0.5       | .302        | [.247, .355] | 2.0%           |
|                         | 1         | .281        | [.231, .330] | 0.0%           |
|                         | 2         | .249        | [.204, .295] | 0.0%           |
| dev / nomic-embed-text  | 0         | .233        | [.168, .299] | 0.7%           |
|                         | 0.25      | .251        | [.188, .315] | 17.5%          |
|                         | 0.3       | .253        | [.189, .315] | 27.6%          |
|                         | 0.4 ★     | .253        | [.189, .315] | 20.4%          |
|                         | 0.5       | .252        | [.189, .314] | 17.8%          |
|                         | 0.6       | .250        | [.188, .312] | 9.9%           |
|                         | 0.75      | .247        | [.186, .308] | 2.6%           |
|                         | 1         | .242        | [.182, .301] | 2.9%           |
|                         | 2         | .221        | [.165, .276] | 0.5%           |
| test / nomic-embed-text | 0         | .314        | [.273, .352] | 0.3%           |
|                         | 0.25 ★    | .334        | [.291, .371] | 85.0%          |
|                         | 0.3       | .332        | [.289, .370] | 14.6%          |
|                         | 0.4       | .326        | [.282, .365] | 0.1%           |
|                         | 0.5       | .319        | [.274, .359] | 0.0%           |
|                         | 0.6       | .311        | [.265, .352] | 0.0%           |
|                         | 0.75      | .299        | [.252, .340] | 0.0%           |
|                         | 1         | .280        | [.228, .325] | 0.0%           |
|                         | 2         | .237        | [.182, .288] | 0.0%           |

| Paired comparison                                  | $P(\lambda_A^*=\lambda_B^*)$ | $P(\lambda_A^*<\lambda_B^*)$ | $P(\lambda_A^*>\lambda_B^*)$ |
|----------------------------------------------------|------------------------------|------------------------------|------------------------------|
| cross-backbone (shared draw)                       | 37.8%                        | 0.4%                         | 61.9%                        |
| same backbone, two independent draws (noise floor) | 43.5%                        | 28.6%                        | 27.9%                        |
| dev vs test, same backbone                         | 46.3%                        | 0.9%                         | 52.8%                        |

Table 14: Embedder ablation: canonical LGS ( $\lambda = 0.5$ ) on the full SummEval set with four sentence-embedder backbones (Ollama). Reports summary-level Spearman  $\rho$  per dimension and the mean across the four. Canonical embedder (used in the headline tables) marked  $\star$ .

| Embedder                                     | Coh  | Con  | Flu  | Rel  | Mean |
|----------------------------------------------|------|------|------|------|------|
| <code>nomic-embed-text</code> (137M) $\star$ | .396 | .227 | .156 | .356 | .284 |
| <code>mxbai-embed-large</code> (335M)        | .304 | .351 | .263 | .266 | .296 |
| <code>all-minilm</code> (23M)                | .291 | .367 | .276 | .259 | .298 |
| <code>bge-m3</code> (567M)                   | .335 | .360 | .251 | .299 | .312 |

## 8. Discussion

### 8.1. Decision-support deployment scenarios

LGS is designed for three decision procedures that reference-based and LLM-judge metrics serve poorly. The first is *online model selection*: a summarisation system serves many candidates per source article (multiple decoding configurations, multiple prompt variants, multiple checkpoints under A/B test), and the operator needs a low-latency, deterministic score to pick the best candidate per request. Reference-based metrics are unavailable because there is no human reference for production inputs; G-Eval’s per-candidate judge call is too expensive and its sampled decoding too non-deterministic to make ranking reproducible across replays. LGS is reference-free, deterministic up to the embedder’s numerical precision, and adds approximately one CPU sentence-embedder forward pass per source and per candidate sentence to the request budget — with the source side amortised across candidates (Section 3.3).

The second is *automated release gating*. A continuous-integration pipeline replays a fixed corpus of source articles through both the previous and candidate model versions and blocks a merge on a significant regression. Such pipelines run at high frequency (per commit, per nightly build) over many thousands of articles; the cost of a judge or fine-tuned-encoder pass per (article, candidate) pair under this load is prohibitive. The reference-free, embedding-only profile of LGS makes the wall-clock overhead negligible, and its single hyperparameter is fixed at deployment time, so the score is a stable signal across runs. Determinism matters here beyond cost: a gate that fires non-deterministically is not a gate.

The third is *reward modelling for reinforcement-learning fine-tuning of summarisation systems*. Such training generates many candidates per article per step, requiring a fast and stable reward signal. LGS produces a deterministic scalar per candidate from a single embedder pass, which is compatible with the iteration speed demanded by these training loops, and the source-side embedding cache is reused across the whole rollout for a given article. We do not claim that LGS is the ideal reward signal — specialised reward models tuned to the target dimension would likely dominate — but it is a viable default when no such specialised model exists, particularly on news-domain corpora where the injected prior matches the underlying distribution.

In all three cases the metric is consumed by an automated procedure that must justify its decision to a human afterwards. The argmax anchors

of Algorithm 1 serve that purpose: a rejected candidate can be reported together with the specific summary sentences that failed to find support in the source and the source sentences that came closest, which neither a whole-text cosine nor a judge’s numeric verdict provides without extra work.

### 8.2. What LGS does and does not claim

We make four claims and explicitly disclaim the stronger versions of each.

- (i) Among the thirteen published, non-trivial baselines in our controlled benchmark, LGS occupies an undominated cost–quality point and strictly dominates six widely used metrics. We do *not* claim Pareto-optimality against the full pool: once reference-free lead-window controls are included, LGS is strictly dominated on the raw mean-correlation axis (Section 6.3), and on the confound-controlled axis it is not statistically distinguishable from the cheapest such control (Section 6.4). Nor do we claim the highest correlation — UniEval is higher on every dimension and we report that directly.
- (ii) The injected lead-bias prior produces a held-out lift on the *mean* Spearman  $\rho$  across the four SUMMEVAL dimensions and on *coherence* specifically; we do *not* claim a held-out per-dimension lift on consistency, fluency, or relevance, where the development-side improvements (+.007, +.011, +.036 respectively) attenuate to within bootstrap noise on test (−.001, +.001, +.003).
- (iii) Imposing some positional prior is supported:  $\lambda=0$  is selected in under 1% of bootstrap resamples on either split. We do *not* claim that  $\lambda^* = 0.5$  is the optimum, universal or otherwise — it wins 17.8% of development resamples, three other grid values win comparable shares, and an unchanged configuration reselects it only 43.5% of the time across two draws. What the protocol returns is the interval  $[0.25, 0.6]$  (Section 7.3). We further claim a *directional* backbone effect — **bge-m3** prefers a weaker prior — but not a quantified shift, and we note that a change of article sample under a fixed backbone produces a directional shift of comparable strength, so the backbone is not isolated as the cause (Section 7.5).
- (iv) The cost figures characterise a single-machine, CPU-only, single-thread deployment target; we do *not* claim that the same ratios hold under GPU serving or batched inference, although the structural difference between one small embedder pass and  $N$  autoregressive decodes is not a property of the hardware.

### 8.3. Practitioner recommendations

For readers considering LGS as a deployed metric, the empirical findings above translate into four operational recommendations. First, *run the*

*calibration sweep.* The optimal  $\lambda$  depends on the backbone (Section 7.5), and reusing a published  $\lambda$  value from a different backbone can degrade the metric below its no-prior baseline. The sweep is cheap — five  $\lambda$  values on a 50-article development split take approximately five minutes on the canonical `nomic-embed-text` backbone, and require no additional embedder passes beyond the first (Section 3.3) — and is the operational price of an explicit structural prior. Second, *decide whether the mean or a single dimension is the deployment target.* The canonical  $\lambda^*=0.5$  optimises the mean across the four SUMMEVAL dimensions; if the deployment cares specifically about coherence or relevance the optimum shifts toward larger  $\lambda$ , and if it cares about consistency or fluency the optimum shifts toward the smallest non-zero  $\lambda$  in the grid (Figure 4). Third, *pick the backbone on the deployment’s cost budget, not on ours.* The backbone ablation (Section 7.4) shows the mean  $\rho$  varies by only .028 across a  $24\times$  parameter range, so a deployment that can afford `bge-m3` gains a slightly higher mean, and one that cannot loses very little by using a 23M-parameter embedder — but the per-dimension profile changes substantially, so the choice should be made against the dimension that matters. Fourth, *re-tune  $\lambda$  when migrating backbones.* A backbone change is not a drop-in replacement; it is a new deployment from the metric’s perspective.

#### 8.4. Limitations

*Domain.* SUMMEVAL contains only CNN/DailyMail news articles. Lead bias is a well-documented structural property of this corpus, and LGS exploits it explicitly through the exponential prior. Domains without a strong leading-sentence structure — scientific abstracting, narrative or dialogue summarisation, long-form question answering — may require a different prior or none at all. We do not claim that the exponential lead-bias formulation transfers; we claim that the *protocol* of selecting and calibrating a structural source-side prior on a held-out split transfers, and that the prior shape itself is a domain-specific design choice.

*G-Eval baseline configuration.* Our G-Eval baseline uses  $N = 3$  independent runs at temperature 0.7 with `qwen2.5:7b-instruct-q4_K_M` and reports the across-run mean as the canonical score. The original paper averages 20 GPT-4 samples; our setup is therefore a budget reproduction. The across-run standard deviation of summary-level Spearman is between .003 and .009 across the four dimensions, well below the half-width of the paired-bootstrap confidence intervals, so judge stochasticity is not a meaningful

confound for the LGS-vs-G-Eval comparisons. We document this explicitly because single-run G-Eval baselines in the literature do not.

*Prior shape.* The lead-bias prior is exponential. Linear, reciprocal, and hard-cutoff alternatives have not been tested; we leave them as future work (Section 8.6).

*Backbone.* The headline numbers use `nomic-embed-text` with  $\lambda^*$  calibrated against it. The ablation in Section 7.4 shows that mean Spearman is qualitatively stable across a  $\sim 24\times$  parameter range, but per-dimension profiles shift; numbers quoted at the per-dimension granularity should be read as backbone-conditional.

### 8.5. Threats to validity

We flag four threats to the validity of the conclusions drawn in this paper that a careful reader should weigh.

*Single-corpus evidence.* All correlation numbers are computed on SUMMEVAL, a single benchmark derived from a single news corpus (CNN/DailyMail). The *methodological* contributions — the article-level held-out split, the cluster-bootstrap discipline, the cost-controlled protocol, and the demonstration that the optimal exponent shifts across backbones — are corpus-independent in principle, but the *numerical* headline correlations transfer to other corpora only to the degree that those corpora share the lead-bias structural property of SUMMEVAL articles. We consider a multi-corpus replication the single most valuable extension of this work (Section 8.6).

*Reproductions versus original-paper numbers.* As discussed in Section 5 and Table 4, our baseline numbers are not the original-paper numbers. They are reproductions under budget-feasible configurations on a single laptop-scale deployment target, which is what makes the cost axis meaningful. Cross-paper number comparisons should therefore be made against the original references, not against our reproductions; our reproductions are internally self-consistent for the cross-baseline comparisons we report, and the Pareto structure of Section 6.3 is a statement about this internally consistent pool.

*Stochasticity of the LLM judge.* Our G-Eval configuration uses  $N=3$  samples at temperature 0.7. The across-run standard deviation of summary-level Spearman is between .003 and .009, which we have argued is small enough not to be a meaningful confound for the LGS-vs-G-Eval comparisons (Section 8.4). A full  $N=20$  reproduction at temperature 1 with a frontier-grade judge would shrink this further and would raise G-Eval’s correlation,

but at a proportionally higher cost, so it would not move G-Eval onto the Pareto frontier unless the correlation gain exceeded UniEval’s.

*Backbone calibration of the score scale.* The cosine similarity values used inside LGS are not calibrated probabilities; they are raw inner products in a learned representation space. The metric is therefore meaningful as a *ranker* of summaries (which is what Spearman measures) but should not be interpreted as a calibrated quality score on an absolute scale.

### 8.6. Future work

Four directions extend the current contribution most directly. First, transfer to non-news corpora, which is where the injected knowledge is most at risk: scientific abstracting (ARXIV, PUBMED, see [48]) and dialogue summarisation (SAMSUM [49], DIALOGSUM) are natural next benchmarks, and the prior shape is the variable we expect to re-tune — or to remove, if the corpus has no positional salience structure. Second, alternative shapes for the source-position prior: linear  $w(i) = \max(0, 1 - \lambda i/n)$ , reciprocal  $w(i) = 1/(1 + \lambda i/n)$ , and a hard cutoff at the first  $k$  sentences. The calibration protocol of Section 4 applies unchanged. Third, richer sources of structural knowledge than position: salience weighting derived from discourse parsing, section headings in structured documents, or entity-centrality signals over the source — all of which fit the same multiplicative source-side slot as  $w(i)$  and can be ablated the same way. Fourth, alternative aggregation operators on the candidate side: low-percentile recall (a hallucination-sensitive variant) and position-weighted top- $k$  recall, evaluated against the diffusion pattern documented in Appendix Appendix B.

## 9. Conclusion

We presented LGS, a reference-free, embedding-only operator for summary quality assessment, defined as  $\text{LGS}(D, C) = \text{mean}_j \max_i w(i) \cos(c_j, s_i)$  with an exponential lead-bias prior  $w(i) = \exp(-\lambda i/n)$  over source-sentence position. The prior is the point of the design: it injects one explicit, auditable piece of discourse-structural domain knowledge into a subsymbolic similarity space, at zero inference cost and with a single interpretable parameter that can be set to zero to remove it exactly.

Measured against thirteen published baselines on SUMMEVAL under one controlled hardware budget, LGS occupies an undominated cost-quality point and strictly dominates six widely used metrics — among them

BERTScore, whose mean summary-level Spearman  $\rho$  it exceeds while running roughly ten times faster and without access to the human reference summary. In the same analysis the LLM-judge baseline is itself dominated by the fine-tuned-encoder baseline — a statement about the budget configurations of Table 4, not about G-Eval with a frontier-grade judge, and one we make only for the reproduction pool defined in Section 5.

That result does not survive the control a reviewer of an earlier version of this paper correctly demanded: a trivial reference-free lead-window heuristic, at roughly three orders of magnitude lower cost, matches or beats LGS’s own raw mean correlation (Section 6.3). We traced this to a confound in the benchmark itself — SUMMEVAL’s human ratings substantially reward verbatim copying from the source, and a parameter-free copy-rate counter alone outranks eleven of the fourteen published metrics in our pool (Section 6.4). Once that confound is controlled for, LGS is no longer statistically distinguishable from the cheapest lead-window control, and it ties rather than beats BERTScore. We regard this as the central limitation of the paper: the sentence-level grounding mechanism LGS is built around does not demonstrably contribute more than a four-line lexical heuristic once the benchmark’s extractiveness confound is removed, and we report that finding in full rather than reframing the metric pool to obscure it. We are explicit that UniEval attains a higher correlation than LGS on every dimension regardless of which axis is used; the claim of this paper, appropriately narrowed, is an undominated position among learned and embedding-based metrics, not a correlation record and not an unqualified Pareto-optimality claim.

The second contribution is methodological and, we believe, transfers beyond this metric. Submitting the hyperparameter *selection procedure* to the same bootstrap that validates the correlations separates two claims that a point estimate conflates. That the prior should be switched on is reproducible: the no-prior setting is selected in under 1% of resamples. Which exponent to switch it to is not: the reported value wins under a fifth of resamples, and an unchanged configuration reselects it less than half the time across two draws of the same articles. A knowledge-informed metric of this kind should therefore be published as a *design plus an obligatory calibration protocol* that returns an interval, not as a fixed numerical recipe — and the width of that interval belongs in the paper, because a practitioner who copies a published exponent is otherwise copying one draw’s answer to an underdetermined question. The calibration is cheap — five  $\lambda$  values on 50 articles, approximately five minutes on a laptop CPU thread, with no addi-

tional embedder passes — and the cost of skipping it is silent miscalibration. The same discipline applies to alternative priors and alternative aggregation operators; Appendix Appendix B records the components we tested and rejected, together with the structural pattern extracted from those negative results.

## Reproducibility

The full pipeline is open-source and deterministic up to the sentence embedder’s numerical precision. The repository (<https://github.com/mikolajsemeniuk/llmbench>) contains a Makefile reproducing every table and figure in this paper end-to-end: `make benchmark` runs all 14 metrics on SUMMEVAL, `make benchmark-ablation` runs the  $\lambda$  sweep on the development and test splits, `make benchmark-embedder-ablation` runs the canonical metric with the four backbones, and `make paper` regenerates every `.tex` table from the corresponding JSON outputs. The selection-uncertainty analysis of Section 7.3 is `make paper-lamdaci`, and the multiple-comparison correction of Section 4.5 is applied inside `cmd/compare`; both read the stored per-sample scores and need neither Ollama nor the model server, so a reviewer can reproduce every statistical claim in this paper without running a single metric. Bootstrap seeds are fixed in `pkg/eval` (seed 42 for cluster bootstrap, seed-derived for paired comparisons). The sentence embedders are pinned by Ollama tag in the Makefile (`nomic-embed-text`, `mxbai-embed-large`, `bge-m3`, `all-minilm`); the G-Eval judge is `qwen2.5:7b-instruct-q4_K_M` under the same Ollama runtime. SUMMEVAL itself is publicly available under the original authors’ release [9]; the JSONL split used for the article-level development/test partition is the SummEval canonical release order, deterministic without a random seed.

## CRedit authorship contribution statement

**Mikolaj Semeniuk:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Agnieszka Jastrzebska:** Conceptualization, Methodology, Validation, Supervision, Project administration, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

## Data availability

All data and code required to reproduce every result in this paper are publicly available. The source code, the reproduction `Makefile`, and the per-sample score files for all fourteen metrics are released at <https://github.com/mikolajsemeniuk/llmbench>. The SUMMEVAL benchmark is publicly distributed by its original authors [9].

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic; Claude Sonnet 5, 2026) in order to improve the language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

## Appendix A. Kendall $\tau$ Correlations

Section 4.4 states that Spearman  $\rho$  is the primary correlation coefficient of this study and Kendall  $\tau$  a secondary check. Tables A.15 and A.16 report the  $\tau$  counterparts of Tables 8 and 10, with the same article-level cluster bootstrap. The two coefficients induce almost the same ordering of the fourteen metrics — the rank correlation between the  $\rho$ -ordering and the  $\tau$ -ordering is .996 on coherence, 1.000 on consistency, .987 on fluency and .996 on relevance, and no metric changes rank by more than two positions on any dimension. The conclusions of Section 7 are therefore invariant to the choice between the two rank coefficients.

Table A.15: Summary-level correlations with human judgment on SummEval.  $\tau$  = Kendall. Brackets give the 95% cluster-bootstrap confidence interval over articles.

| Metric       | Coh                 | Con                 | Flu                | Rel               |
|--------------|---------------------|---------------------|--------------------|-------------------|
| BLEU         | .072 [.024, .115]   | .107 [.065, .147]   | .051 [.005, .097]  | .148 [.094, .203] |
| ROUGE-L      | .115 [.067, .162]   | .110 [.062, .154]   | .083 [.036, .134]  | .144 [.084, .202] |
| ChrF         | .112 [.065, .160]   | .137 [.101, .174]   | .086 [.043, .132]  | .210 [.158, .264] |
| METEOR       | .139 [.100, .176]   | .105 [.064, .143]   | .101 [.051, .151]  | .131 [.078, .179] |
| SMART-String | -.004 [-.051, .043] | .087 [.043, .130]   | .081 [.037, .123]  | .088 [.034, .143] |
| EmbedScorer  | .197 [.153, .238]   | .044 [-.004, .088]  | .031 [-.018, .081] | .229 [.179, .279] |
| BERTScore    | .269 [.225, .307]   | .085 [.038, .128]   | .110 [.063, .156]  | .270 [.224, .318] |
| MoverScore   | .106 [.059, .150]   | -.020 [-.066, .026] | .009 [-.045, .057] | .072 [.016, .125] |
| SMART-Model  | .233 [.191, .273]   | .074 [.029, .116]   | .052 [.004, .103]  | .229 [.182, .277] |
| BARTScore    | .117 [.075, .161]   | .092 [.046, .137]   | .083 [.033, .132]  | .195 [.141, .252] |
| GPTScore     | .256 [.218, .293]   | .166 [.119, .216]   | .142 [.096, .194]  | .221 [.175, .264] |
| UniEval      | .408 [.378, .437]   | .323 [.293, .353]   | .352 [.317, .385]  | .334 [.293, .370] |
| G-Eval       | .318 [.276, .361]   | .448 [.410, .484]   | .131 [.094, .165]  | .345 [.293, .398] |
| LGS          | .285 [.243, .326]   | .179 [.137, .219]   | .120 [.076, .169]  | .257 [.216, .298] |

Table A.16: System-level correlations with human judgment on SummEval.  $\tau$  = Kendall. Brackets give the 95% cluster-bootstrap confidence interval over articles.

| Metric       | Coh                 | Con                  | Flu                 | Rel                 |
|--------------|---------------------|----------------------|---------------------|---------------------|
| BLEU         | .183 [-.276, .607]  | .017 [-.407, .459]   | .283 [-.228, .691]  | .350 [-.146, .709]  |
| ROUGE-L      | .217 [-.243, .636]  | -.017 [-.477, .424]  | .250 [-.183, .648]  | .317 [-.143, .712]  |
| ChrF         | .350 [-.045, .673]  | .617 [.351, .852]    | .617 [.357, .835]   | .617 [.375, .804]   |
| METEOR       | .000 [-.472, .404]  | -.200 [-.568, .264]  | .033 [-.439, .523]  | .067 [-.439, .486]  |
| SMART-String | -.150 [-.655, .297] | -.117 [-.469, .268]  | .117 [-.299, .535]  | .150 [-.386, .593]  |
| EmbedScorer  | .350 [.009, .646]   | .283 [-.091, .630]   | .183 [-.232, .540]  | .217 [-.171, .596]  |
| BERTScore    | .617 [.254, .887]   | -.017 [-.473, .460]  | .250 [-.232, .702]  | .383 [-.086, .796]  |
| MoverScore   | .083 [-.416, .541]  | -.450 [-.764, -.045] | -.283 [-.707, .174] | -.150 [-.655, .345] |
| SMART-Model  | .533 [.164, .802]   | .467 [.055, .790]    | .400 [-.089, .784]  | .467 [-.027, .874]  |
| BARTScore    | .367 [-.064, .694]  | .633 [.382, .870]    | .567 [.315, .784]   | .600 [.327, .835]   |
| GPTScore     | .500 [.053, .842]   | .400 [-.018, .762]   | .300 [-.236, .770]  | .367 [-.141, .852]  |
| UniEval      | .650 [.298, .911]   | .367 [-.057, .750]   | .583 [.252, .836]   | .633 [.321, .844]   |
| G-Eval       | .400 [-.018, .752]  | .745 [.491, .909]    | .226 [-.271, .661]  | .650 [.256, .927]   |
| LGS          | .467 [.009, .892]   | .467 [.135, .735]    | .400 [-.088, .807]  | .433 [-.035, .861]  |

## Appendix B. Extension Log: Components Tested and Rejected

The canonical LGS reported in this paper has a single tunable hyperparameter,  $\lambda$ . This minimal design is the result of a sequence of ablations conducted during calibration; each rejected component below was tested under the same held-out protocol of Section 4 (sweep on the 50-article development split, mean Spearman  $\rho$  across the four SUMMEVAL dimensions as the selection criterion, verification on the 50-article test split). We document them here so that readers can see what was tried, why it was rejected, and what general lesson the rejection encoded.

Table B.17 summarises the development-split mean Spearman  $\rho$  at the optimum of each rejected component’s grid, against the no-prior recall baseline ( $\rho=.233$ ) and the accepted lead-bias prior ( $\rho=.252$  at  $\lambda^*=0.5$ ).

Table B.17: LGS components tested during calibration. Development-split mean Spearman  $\rho$  averaged across the four SUMMEVAL dimensions on the 50-article development split. Only the lead-bias prior was retained in the canonical metric.

| Component                           | Form                                                                        | Dev $\rho$ at optimum                  | Verdict                                                                         |
|-------------------------------------|-----------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------|
| Recall baseline                     | $\text{mean}_j \max_i \cos(c_j, s_i)$                                       | .233                                   | reference point.                                                                |
| Lead-bias prior $\lambda$           | $\text{mean}_j \max_i w(i) \cos(c_j, s_i),$<br>$w(i) = \exp(-\lambda i/n)$  | <b>.252</b> ( $\lambda^*=0.5$ )        | <b>accepted:</b> only component to lift the development mean over the baseline. |
| Legacy precision                    | $F_\beta$ $F_\beta(R, P)$ with $P$ over top-30% source by degree centrality | .228 ( $\beta=2$ , $\text{frac}=.30$ ) | rejected: aids relevance only by $\sim+.02$ , loses 3/4 dimensions.             |
| Anchor coverage $C^\alpha$          | $R \cdot C^\alpha$ , $C= \text{distinct argmax}_i /m$                       | .229 ( $\alpha=.25$ )                  | rejected: optimum at $\alpha=0$ (no coverage).                                  |
| Within-summary diversity $D^\gamma$ | $R \cdot D^\gamma$ , $D=1-\text{mean}_{j<k} \cos(c_j, c_k)$                 | .209 ( $\gamma=.10$ )                  | rejected: monotonically harmful for $\gamma>0$ .                                |
| Top- $k$ recall                     | $\text{mean}_j \frac{1}{k} \sum_{\text{top-}k} \cos(c_j, s_i)$              | .213 ( $k=2$ )                         | rejected: monotonically harmful for $k>1$ .                                     |

*Pattern across rejections.* Three of the four rejected components — coverage, top- $k$  recall, and within-summary diversity — share a structural feature: each multiplies the recall term by a value in  $[0, 1]$  that *diffuses* the matching signal across multiple candidate–source pairings. The empirical pattern

is consistent: each diffusion-style component improves relevance marginally while degrading coherence, consistency, and fluency more than it helps. We interpret this as evidence that a sharply localised candidate-to-source grounding (single best match, no penalty) is the strongest signal for SUMMEVAL on three of four dimensions, and that modifications which disperse this localisation hurt overall correlation.

The accepted component (the lead-bias prior) does *not* diffuse. It is a source-side structural prior that concentrates weight on high-information positions *before* the argmax, without spreading the candidate-side aggregation. This is consistent with the diffusion hypothesis: priors that reshape the source without spreading the candidate-side aggregator are the operations most likely to help.

*Implication for future variants.* Future extensions of LGS proposed under the same protocol should be evaluated against this pattern. A new component that multiplies recall by a candidate-side coverage or diversity term is unlikely to lift the development mean unless its formulation specifically avoids the diffusion problem documented here. Source-side priors (alternative decay shapes, salience-based weighting, structural weighting derived from discourse parsing) are the higher-prior direction for future work; we list specific candidates in Section 8.6.

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